CONSTRUCTOR UNIVERSITY

Resilience and Development in Sub-Saharan Africa: Analyzing the Socioeconomic Impact of HIV/AIDS (1990–2023)

Master Thesis

Data Science for Society and Business

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Abstract

This thesis, Resilience and Development in Sub-Saharan Africa: Analysing the Socioeconomic Impact of HIV/AIDS (1990–2023), analyses the influence of the HIV/AIDS pandemic on Sub-Saharan Africa's developmental trajectories. Utilizing panel data for over two decades, the research analyses the influence of HIV prevalence on GDP growth alongside other socioeconomic indicators. At the heart of the analysis is building a multidimensional HIV/AIDS Resilience Index which includes health indicators (e.g., life expectancy and ART coverage), education, and social solidarity to capture each country's capacity to contain the epidemic. The study defines three time periods: T1 (the initial shock phase), T2 (the middle-pandemic phase with scale-up of interventions), and T3 (the recovery and post-period). Empirical estimates derived from fixed-effects panel regressions indicate that while higher HIV prevalence first constrained GDP growth, subsequent rises in ART coverage, education spending, and decentralization benefited improved health and economic performance. Policy recommendations and directions for future research are outlined, with specific emphasis on consolidation of health, education, and governance reforms to foster long-term preparedness against health crises.

Executive Summary

This thesis examines the socioeconomic impact of HIV/AIDS in Sub-Saharan Africa between 1990 and 2023. It adopts a multi-dimensional approach by not just examining conventional economic indicators such as GDP growth and life expectancy but also the effects of education, social cohesion, migration, and aid flows into a novel HIV/AIDS Resilience Index. It is structured chronologically into three phases:

T1 – The Pandemic's Onset: A time of virulent shock with abrupt spikes in HIV prevalence, plummeting life expectancy, and tepid policy response.

T2 – Mid-Pandemic Period: A tipping point at which increased ART programs (augmented by global initiatives like PEPFAR) and investments in schools and health facilities begin to mitigate the adverse effects.

T3 – The Recovery and Aftermath: A period of sustained structural recovery, where improved health outcomes, improved human capital, and stronger social cohesion come to drive modest economic growth. Botswana, South Africa, Ethiopia, Kenya, and Rwanda were the resilience leaders, largely due to their early investment in quality public health systems, education reform, and decentralized government. On the other hand, nations such as Zimbabwe, Lesotho, and Mozambique recovered more gradually due to the problem of governance and policy discontinuities. Policy implications and future research avenues are the thesis's conclusion, supporting the necessity for integrated, multi-dimensional solutions to address complex health emergencies.

CHAPTER 1- Introduction

1. Introduction

1.1 Context

HIV/AIDS has had a devastating effect on the health and socioeconomic condition in Sub-Saharan Africa. The region has borne a disproportionate share of the global HIV/AIDS pandemic since the early 1980s, causing millions of premature deaths, widespread social disruption, and enormous economic costs. Not only has the disease devastated individual lives and communities, but it has also drastically altered national development trajectories. Labour productivity has declined, health systems have been overwhelmed, and basic sectors such as education, agriculture, and mining have suffered permanent losses.

The initial studies, e.g., Dixon, McDonald, and Roberts (2001, 2002), focused on the macroeconomic dimensions of the crisis particularly the reverse correlation between HIV prevalence and GDP growth. However,
focusing on narrowly economic indicators provides only part of the picture. The HIV/AIDS epidemic also undermines the social cohesion of affected societies, draining human capital, eroding institutions, and exacerbating inequality.

In the last few decades, there has been a radical change. Greater access to antiretroviral therapy (ART) maintained by international programs such as PEPFAR and the Global Fund has contributed to rising life expectancy and gradual stabilization in some countries. There has been uneven progress across the region, however, and the mechanisms behind economic and social resilience in the presence of long-term health crises remain inadequately understood.

This thesis takes a multi-dimensional and longitudinal approach to the socioeconomic impact of HIV/AIDS in Sub-Saharan Africa from 1990 to 2023. Moving beyond traditional economic indicators, it introduces a novel HIV/AIDS Resilience
Index that integrates indicators of education, governance, health infrastructure, migration, and social cohesion.

1.2 Problem Statement

Although economists and public health officials have studied the economic burden of HIV/AIDS, much remains to be understood about how countries effectively cope with or struggle against this burden. Most research has been aimed at macroeconomic indicators, ignoring important social factors. Precisely, the potential role of education, Government spending and human capital in a country's economic response to HIV/AIDS has not yet been investigated. These are the regions which determine the way individuals adapt, earn their livings, and contribute to their economies, particularly in the long run.

Then there's the matter of migration and aid, how do these flows of people and money intersect with HIV/AIDS challenges? Do they help some countries become more capable

of managing? These are the kinds of questions this thesis tries to explore, with newer information and a wider range of analytical outlook.

1.3 Research Questions

1. How have HIV/AIDS impacted economic development paths in Sub-Saharan Africa between 1990and 2023?

2. How do education levels, human capital, and social cohesion affect the economic resilience of nations to HIV/AIDS?

3. How could Sub-Saharan African countries be classified with respect to their abilities to manage the economic aspects of HIV/AIDS pandemic?

1.4 Objectives

• To holistically assess the impact of HIV/AIDS-related deaths and loss of human capital on economic development over the course fo 20 years.

• To build a resilience measure for capturing the degree to which countries have coped with the economic effects of HIV/AIDS, using indicators like health, education, and social trust.

• To examine the relationship between this index and significant development results like GDP per capita growth and gains in life expectancy or school enrollment.

1.5 Contribution

This thesis builds on the foundational work of scholars like Dixon, McDonald, Roberts, and Haacker, but takes a step further. It introduces newer variables and frameworks, such as the role of government spending and education, which were largely absent in earlier analyses. The aim is to provide a richer picture of how countries respond to health crises and not just in terms of economic loss, but in terms of social resilience and capacity to recover. The thesis also uses panel data covering over two decades, offering a long-term view that aligns with the sustainable development agenda.

Chapter 2 – Literature Review

2. Literature Review

2.1 Initial Economic Research on HIV/AIDS

The economic contribution of HIV/AIDS began to gain serious attention during the 1990s when the scale of the epidemic came into perspective. Scholars like Over (1992), Cuddington (1993), and Bloom and Mahal (1997) examined how the disease would reduce labour supply, increase healthcare costs, and decay productivity. Most of these early studies used the models of growth that tried to approximate future circumstances and focused primarily on GDP and labour dynamics. Some of the findings were that HIV/AIDS was capable of slowing long-run growth by decreasing the number of economically active, working-age people and diverting resources into treatment and care from investment.

2.2 The Macroeconomic Approach

Dixon, McDonald, and Roberts (2001, 2002) made a valuable contribution using panel data to measure the direct impact of HIV prevalence on economic growth among African countries. Their findings showed a consistent negative relationship between HIV/AIDS and GDP per capita growth. McDonald and Roberts (2006) used a human capital approach and showed that health, like education, plays a central role in leading development. These investigations underlined that HIV/AIDS diseases influence the entire economy not only by reducing the workforce, but also by eroding institutions and government budgets.

2.3 Human Capital and Education

Education constitutes an important element of human capital and has been thought to be a driving force behind economic growth. However, in the HIV/AIDS scenario, the connection between education and growth becomes complex. Some studies (e.g., Bonnel, 2000) were able to determine that HIV/AIDS not only reduces labour productivity but also distorts education systems by affecting teachers and students. Furthermore, orphanhood caused by the epidemic has long-term implications for schooling and educational achievement, especially in poor families. Despite education being commonly used as a control variable within growth regressions, there have been limited studies on how it might shield against the economic effects of HIV/AIDS.

2.4 Migration, Remittances, and Aid

The other important but not commonly discussed factor is the impact of migration and money transfers. Most of Sub-Saharan Africa moves to cities or other countries to work and sends their earnings back home. Remittances allow families to buy healthcare, education, and food

and thereby help indirectly with economic stability. There are arguments (Haacker, 2011) that migration will yield both positive and negative implications for economic growth and HIV transmission. There has also been a large part played by aid, specifically HIV/AIDS funding. However, the long-term structural effect of these flows on development remains contentious.

2.5 Research Gaps

There remain research gaps in spite of a growing literature. There is still scarce understanding of the interaction between education, government spending, and migration and the economic effect of HIV/AIDS. Most existing models are able to take into account short-term effects or a limited number of variables, like GDP growth or health spending. Few studies consider a more consolidated, multi-aspect view, including human and social capital. This thesis attempts to fill this gap by utilizing economic, health, and social measures of resilience and development over a 23-year timeframe.

2.6 Long-term changes in the demography and inter-generational effect

Early work focused on the immediate economic impacts of HIV/AIDS, but more recent studies have placed emphasis on long-term demographic effects and inter-generational transmission. One of the most tangible effects of the epidemic has been the rise in orphanhood, its effects on educational attainment, psychological growth, and future job opportunities. This was found by working with Sub-Saharan African orphans by Case, Abelidinger, and Paxson (2004), who found that orphans had lower school attendance and were more likely to fall behind class levels. This reduction in human capital investment perpetuates poverty cycles and limits the economic opportunities of future generations. HIV/AIDS also alters population dynamics by increasing adult mortality, decreasing fertility rates, and elevating dependency ratios, most significantly in high-prevalence nations with low coverage of ART in the initial years. These alterations place additional pressure on extended family units and social protection systems, which may already be under pressure. The demographic driver also affects the long-term savings propensity, supply of labour, and national planning, but these processes are under-accounted for in macroeconomic models. Taking demographic transitions into consideration when analysing resilience enables us to view things more comprehensively regarding how the epidemic influences not just the existing situation but the course of development of societies over many decades.

2.7 Political Economy of Aid and Health Governance

The political economy of HIV/AIDS aid and the integration into national health governance has also become an important area of study. While urgently needed targeted international aid especially from PEPFAR, the Global Fund, and UNAIDS has been instrumental in getting more individuals on ART, scholars have raised questions about the long-term implications of such aid on institutional development and policy autonomy. Easterly (2006) and Moyo (2009) contend that dependence on foreign aid is likely to skew accountability mechanisms, diminish incentives for mobilizing domestic revenue, and undermine policy consistency. However,

others, such as Kenny and Sandefur (2019), propose that well-articulated aid interventions can enhance national capacity if they are responsive to domestic priorities and complemented with strong monitoring mechanisms. In the HIV/AIDS case, differential country performance is a result of differences in the quality of governance, bureaucratic effectiveness, and absorptive capacity. Rwanda, for example, has been praised for using donor funds to build well-integrated health systems, while others have struggled with disjointed service delivery and inadequate coordination. The arguments also underpin the importance of thinking about resilience not merely in terms of epidemiological or economic resilience but also in terms of institutional adjustment, policy learning, and strategic stewardship of health agendas. Understanding how aid reacts to governance arrangements is central to assessing whether external inputs facilitate or suppress long-term development in the wake of health shocks.

Chapter 3 - Theoretical Framework

3. Theoretical Framework

3.1 Overview

This chapter presents the theoretical framework that guides the analysis of how HIV/AIDS affects economic development in Sub-Saharan Africa. It brings together several key ideas from economic growth theory, human capital theory, Government spending and social cohesion . These concepts help explain why some countries may cope better than others in the face of the epidemic, and how education and social factors can influence development outcomes.

3.2 Health and Economic Growth Theory

Traditional models of growth, such as the **Solow model**, emphasize technology, labor, and capital as the primary drivers of long-run economic growth. However, later growth models incorporated human capital—including **education and health**—to better explain disparities in growth rates across countries (Barro, 1991; Mankiw, Romer, & Weil, 1992). Poor health, particularly in the form of widespread diseases like **HIV/AIDS**, reduces the working-age population, diminishes productivity, and places a financial strain on public resources due to increased healthcare spending. These effects can hamper economic growth and disrupt both private and public investment (Bloom & Canning, 2000; Haacker, 2004)

Real-world examples in Sub-Saharan Africa illustrate long-term economic effects of the HIV/AIDS epidemic. For instance, in Botswana one of the highest HIV prevalence rates countries economic growths in the early 2000s slowed significantly despite the country's well-developed institutions and diamond wealth. Haacker (2004) estimated that GDP per capita annual growth rate in countries with adult HIV prevalence of over 20% can decline by up to 1.5 percentage points.

Similarly, Dixon, McDonald, and Roberts (2001) also found that a 10% rise in HIV prevalence equated to a reduction of 0.6 percentage points in the African countries' rate of growth of per capita GDP. This is not just the instance of falling labour supply but of falling investment in education and infrastructure due to the pressure on health systems.

In **Malawi**, high HIV/AIDS mortality rates during the 1990s and early 2000s disrupted agricultural production systems, leading to widespread **food insecurity** and the erosion of rural livelihoods. These examples underscore how severely health shocks can influence development trajectories, reinforcing the importance of incorporating **health variables** into macroeconomic analysis (Whiteside, 2002; Gillespie & Kadiyala, 2005; Haacker, 2004).

Here, HIV/AIDS is thought to be an unfavourable health shock that influences growth both directly and indirectly. It reduces labour supply, influences school enrolment, and increases dependency ratios. Understanding this relationship allows one to assess how deeply the epidemic has shaped economic trajectories within the region.

3.3 Human Capital and Education

As a matter of fact, human capital theory identifies knowledge, skills, and health of people as the key for economic performance. Countries that have more education and with better health prove to grow faster and to prove to absorb shocks much smoother. When it comes to HIV/AIDS, education has both roles to play improving productivity and increasing awareness and prevention.

This perspective is also supported by empirical evidence. For example, Bloom, Canning, and Sevilla (2004) found that an extra year of schooling can increase the individual wage as much as 10% while more educated populations are more flexible and inventive at the national level. In Sub-Saharan Africa, the only country that has massively invested in human capital in the last few decades is Mauritius. Despite being a small island nation, Mauritius attained long-term economic growth due in part to its focus on expanding access to education and training. This investment paid off, especially during periods of economic uncertainty, since the population had the skills required to move into new sectors like tourism and financial services.

On the other hand, countries with weaker education systems have struggled to recover from the disruptions caused by the HIV/AIDS crisis. For instance, **Lesotho** and **Eswatini**, both characterized by high HIV prevalence and limited access to higher education, have experienced slower economic growth in comparison to more education-focused neighboring countries. These examples illustrate that while education alone cannot fully mitigate the effects of a health crisis, it plays a crucial role in strengthening national resilience (Bloom & Mahal, 1997; Haacker, 2004).

Haacker (2011) also noted that the more educated individuals are, the more likely they are to engage in public health programs and understand preventive behaviour, hence reducing infection rates over time. Education, thus, is not only a productivity booster, but it also plays a pivotal role in shaping the response of societies to HIV/AIDS.

Bachelor's-level education is used in this thesis as a proxy for superior human capital. Though it fails to show the whole picture, it helps capture the way access to tertiary education may influence national resilience to health crises.

3.4 Migration and Aid

Migration and transfer of finances like foreign aid and remittances are also important. They aren't usually included in conventional growth models but do factor in the African context. Migration lowers domestic labour supply but brings in resources in the way of remittances. Foreign aid dedicated to HIV/AIDS initiatives, for example, would enhance health indicators and stabilize government finances.

Evidence from the real world suggests that remittances are increasingly becoming a major source of income in the majority of African countries. For example, in 2022, remittances accounted for over 6% of GDP for both Nigeria and Senegal, according to World Bank data. In the majority of instances, the flow allows households to access healthcare, education, and food, acting as a buffer during economic or health crises. In nations such as Zimbabwe, remittances were especially crucial under episodes of political turmoil and the disintegration of public health, such as during HIV/AIDS surges in the early 2000s.

Foreign aid has also been the focal point of funding HIV/AIDS programs in the country. With initiatives like PEPFAR (the U.S. President's Emergency Plan for AIDS Relief), billions of dollars have flowed into African healthcare systems, funding antiretroviral therapy, training for healthcare staff, and facility development. While others argue that aid may be capable of fostering dependency, a study conducted by Mishra and Newhouse (2009) found that certain HIV/AIDS aid had been associated with reduced prevalence rates and increases in life expectancy.

Migration and aid therefore are both a lifeline and a source of complexity. They can reduce some of the negative effects of HIV/AIDS but also introduce new issues like brain drain or episodic funding. The thesis has migration and aid both as prominent external drivers for resilience and long-term development outcomes in the region.

This model incorporates migration and assistance as extraneous forces that determine countries' ability to cope with the economic impact of HIV/AIDS.

3.5 Conceptual Model

This thesis proposes a Human Capital Pathway through which HIV/AIDS prevalence influences economic growth. The pathway is shown in Figure 1 below and emphasizes the central role of human capital—here defined as the health, skills, and education of a population—as the primary mechanism linking the HIV/AIDS epidemic to long-run economic outcomes.

HIV/AIDS Prevalence → Human Capital → Economic Development

This pathway is hypothesis-driven. It is supposed that higher HIV/AIDS prevalence reduces human capital (through increased morbidity, mortality, and caregiving requirements), which in turn slows economic development.

Modifiers (Exploratory):

These contextual variables are supposed to condition the direction and strength of the main pathway. Their specific functions are exploratory and may vary by setting.

Education: More tertiary education will be anticipated to buffer the negative effect of HIV/AIDS through enhanced awareness, prevention, and adaptability—enhancing resilience in society.

Government Spending on ART (Antiretroviral Therapy): Effective public spending on ART initiatives can be anticipated to mitigate the influence of the epidemic on human capital by prolonging productive lives and averting transmission.

Migration and Aid Flows: These can operate in complex ways—either alleviating pressure through remittances and external support or compounding brain drain and dependency.

Hypotheses and Exploratory Components

Hypothesis-Driven Components:

H1: Higher HIV/AIDS prevalence is associated with lower human capital.

H2: Lower levels of human capital lead to lower economic development.

H3: ART and education spending moderate (reduce) the negative impact of HIV/AIDS on human capital.

Exploratory Components:

The interaction effects of aid and migration on resilience.

The combined influence of all modifiers on recovery trajectories.

Goal Alignment: Toward a Resilience Index

In line with the thesis objective, this framework also guides the construction of a Resilience Index a composite indicator aggregating:

1. Human capital indicators (e.g., life expectancy, literacy),
2. Policy responsiveness (e.g., health expenditure, ART coverage),
3. Adaptive capacity (e.g., demographic trends, access to education).

The Resilience Index will be both an analytical tool and a policy-relevant outcome to benchmark countries' capacity to absorb and recover from the HIV/AIDS shock.

3.6 Hypotheses

Based on this framework, the following hypotheses are proposed:

1. Higher HIV/AIDS prevalence is associated with lower GDP growth in Sub-Saharan African countries.
2. Countries with higher levels of education experience less economic decline during the HIV/AIDS epidemic.
3. Migration and aid flows influence national resilience to HIV/AIDS-related shocks.

The next chapter outlines how the data are collected and how these relationships are tested using panel data analysis.

Chapter 4 - Data and Methodology

4. Data and Methodology

4.1 Data Sources

This thesis uses panel data covering Sub-Saharan African countries from 2000 to 2023. The dataset was compiled from several reputable international sources:

- **World Bank Development Indicators**: for GDP growth (annual %), GDP per capita (current US$), life expectancy at birth, and total population.
- **World Bank Education Statistics**: for the percentage of the population aged 25+ with at least a bachelor's degree (as a proxy for higher educational attainment).
- **UNAIDS**: for adult HIV prevalence rates and antiretroviral treatment (ART) coverage..
- **World Bank Migration and Remittances Dataset**: for net migration rates and remittances received as a percentage of GDP.

All datasets were harmonized and cleaned to ensure consistency across countries and years.

4.2 Key Variables

Dependent Variable

- **GDP growth (annual %)**: This is the primary measure of economic development used in the regression analysis.

Independent Variables

- **HIV Prevalence Rate**: Percentage of adults aged 15–49 living with HIV, used to represent the burden of the epidemic.
- **Life Expectancy at Birth**: Proxy for general health conditions and improvements over time.
- **Education (% with Bachelor's Degree)**: Proxy for advanced human capital and educational attainment.
- **Government Spending on Health**
- **Migration Rate and Remittances (% of GDP)**: Represent external support and household resilience.

All variables were converted into panel format with consistent time frames and missing values addressed through interpolation where feasible.

4.3 HIV/AIDS Resilience Index

To capture how effectively each country managed the HIV/AIDS crisis, a composite **HIV/AIDS Resilience Index** was created using four components:

1. **Change in Life Expectancy** (–2023)
2. **Increase in Bachelor's-Level Education** (% change over time)
3. **ART Coverage** (% of eligible population receiving treatment)
4. **Government Health Expenditure (% of GDP):**

Each component was normalized to a 0–1 scale and weighted equally. The index offers a comparative measure of resilience across countries and is used to explore relationships with GDP growth.

4.4 Econometric Approach

The analysis uses fixed-effects panel regression to estimate the impact of HIV/AIDS and mediating variables on economic development. The fixed-effects model controls for unobserved, time-invariant characteristics of each country.

The model specification is:

GDP_growth_it = β0 + β1(HIV_it) + β2(Education_it) + β3(LifeExpectancy_it) + β4(SocialCohesion_it) + β5(Migration_it) + β6(Remittances_it) + μ_i + ε_it

Where:

- *i* = country
- *t* = year
- *μ_i* = country-specific fixed effect
- *ε_it* = error term

Interaction terms are also included to test whether education and government spending reduce the negative effect of HIV prevalence on economic growth.

Here's a breakdown of what each component means:

- **GDP_growth_it**: This is the annual percentage change in GDP for country *i* in year *t*. It's the main outcome we want to explain.
- **β0**: This is the intercept — the baseline level of GDP growth when all other variables are zero.
- **HIV_it**: The HIV prevalence rate in a country — a higher rate is expected to negatively affect GDP growth because of impacts on the labor force and health system.
- **Education_it**: The percentage of the population with at least a bachelor's degree. Higher education is assumed to strengthen a country's resilience by improving productivity and health awareness.

- **LifeExpectancy_it**: Life expectancy at birth, representing overall population health. Longer life expectancy generally signals better development conditions.
- **Migration_it**: The net migration rate, capturing how the movement of people may affect the workforce and household stability.
- **Remittances_it**: The value of remittances received (as a % of GDP). These funds can support household resilience and reduce pressure on public services.
- **μ_i**: These are fixed effects — they control for unique characteristics of each country that do not change over time, such as geography or cultural factors.
- **ε_it**: The error term — the portion of GDP growth not explained by the model.

4.5 Limitations

There are some data limitations to note:

- **Unbalanced Panel**: Some variables contain missing values for certain countries or time periods.
- **Equal Weighting Assumptions**: The resilience index components are weighted equally, which may oversimplify the variation in their real-world influence.
- **Endogeneity**: There may be reverse causality between some independent variables and GDP growth, though fixed effects help mitigate this concern.

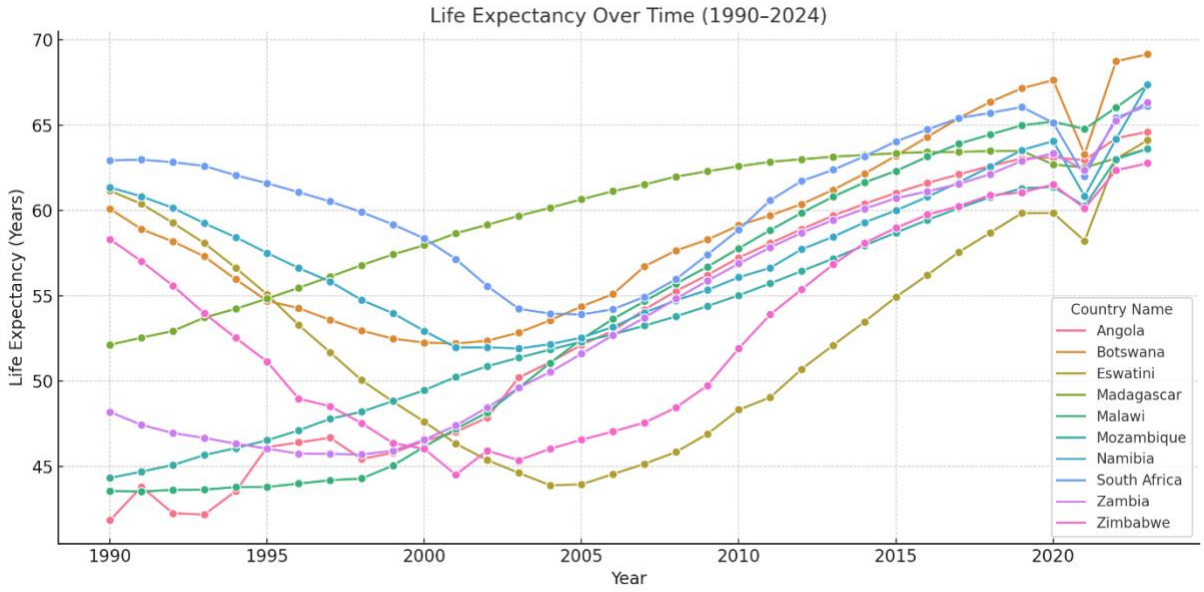
Despite these limitations, the dataset provides a rich basis for analysis and supports robust, theory-driven testing of the hypotheses presented in Chapter 3.

Chapter 5: Exploratory Data Analysis (EDA) and Visualizations

5. Data Visualization and Insight Exploration

This chapter presents a comprehensive set of visualizations created during the exploratory phase of the research. The primary goal of this section is to provide an overview of the data, identify potential patterns, and highlight areas of interest that informed later stages of analysis. While not all visualizations shown here directly support the central arguments made in subsequent chapters, they represent important steps in the data understanding process. The inclusion of both relevant and tangential findings is intentional, offering transparency into the analytical journey. Readers are encouraged to focus particularly on Figures 5.1, 5.2, 5.4 and 5.9, which most strongly relate to the thesis' core claims.

5.1 An analysis of life expectancy



Analysis of life expectancy trends in the selected ten Southern African countries between the years 1990 and 2023 shows overall improvement in the health outcome since the previous three decades. The graphical information shows a consistent increase in life expectancy in all countries but at varying rates. In South Africa, Namibia, and Botswana, life expectancy, having plummeted in the late 1990s and early 2000s due to the HIV/AIDS pandemic, regained ground steadily since 2004, coinciding with the bulk introduction of antiretroviral therapy (ART). This is consistent with evidence gathered by Bor, Herbst, Newell, and Bärnighausen (2013), which reported that ART programs in Southern Africa not only contributed to a decline in HIV-related mortality but also to broader improvements in public health outcomes, including gains in life expectancy.

Eswatini and Lesotho, which initially reported the largest HIV prevalence rates in the world, experienced more precipitous declines in life expectancy in the early 2000s. However, following intense public health campaigns and international aid programs such as PEPFAR (President's Emergency Plan for AIDS Relief), these countries also recorded strong recoveries

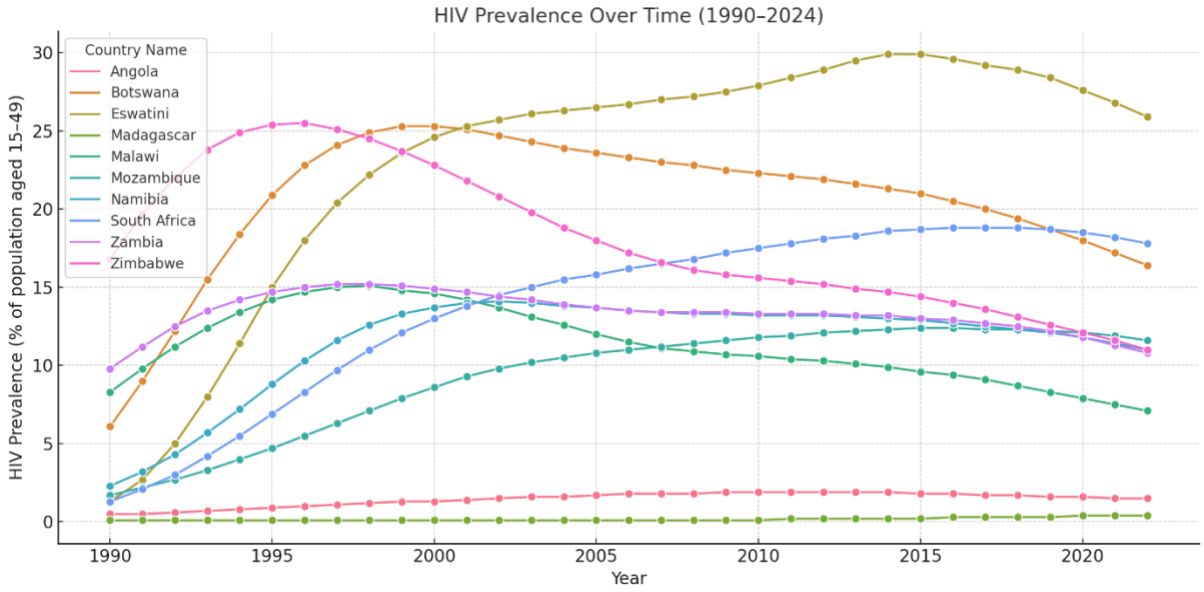
in the mid-2010s. Johnson et al. (2015) noted that effective ART coverage and large-scale HIV prevention programs were major drivers that reversed previous downward trends.

Despite overall progress, there are still lingering disparities between the nations. Those nations with earlier and more widespread ART rollouts, such as Botswana and Namibia, have a steeper slope of life expectancy compared to nations like Mozambique and Malawi, where constraints in the health system and socioeconomic challenges slowed recovery. This supports the argument that healthcare outcomes are not solely dictated by biomedical interventions but also by broader systemic determinants such as education, governance, and public trust (Haacker, 2004; Campbell, 2003).

Secondly, the updated data show that gains in life expectancy were not always linear. Some countries experienced short-term flattening or minor reversals between 2008 and 2010, largely because of worldwide financial stress on healthcare budgets and program growth. However, the overall positive trend demonstrates the resilience of the region's health systems and suggests the pioneering impact of sustained international health investments, improved national policy, and increased popular engagement in health programs.

Together, these achieved increases in life expectancy are a core element of the resilience model presented in this thesis. They affirm that, while HIV/AIDS posed an unprecedented threat to development, concerted health interventions alongside broader social and economic investments have allowed many Southern African countries to recover and establish firmer foundations for ongoing development.

5.2 Trend analysis of HIV prevalence from 1990 to 2024



The pattern from 1990 to 2024 in HIV prevalence within the selected Southern African countries displays both the severity of the epidemic and the measurable impact of health interventions over the years. In the early 2000s, countries such as Eswatini, Lesotho, and Botswana had among the highest rates of HIV prevalence globally, with adult prevalence levels exceeding 25% in a few cases (UNAIDS, 2006). As reflected in the new visual data, HIV prevalence was high and relatively stable across this period, underscoring the severity of the epidemic and the delay between transmission dynamics and public health response.

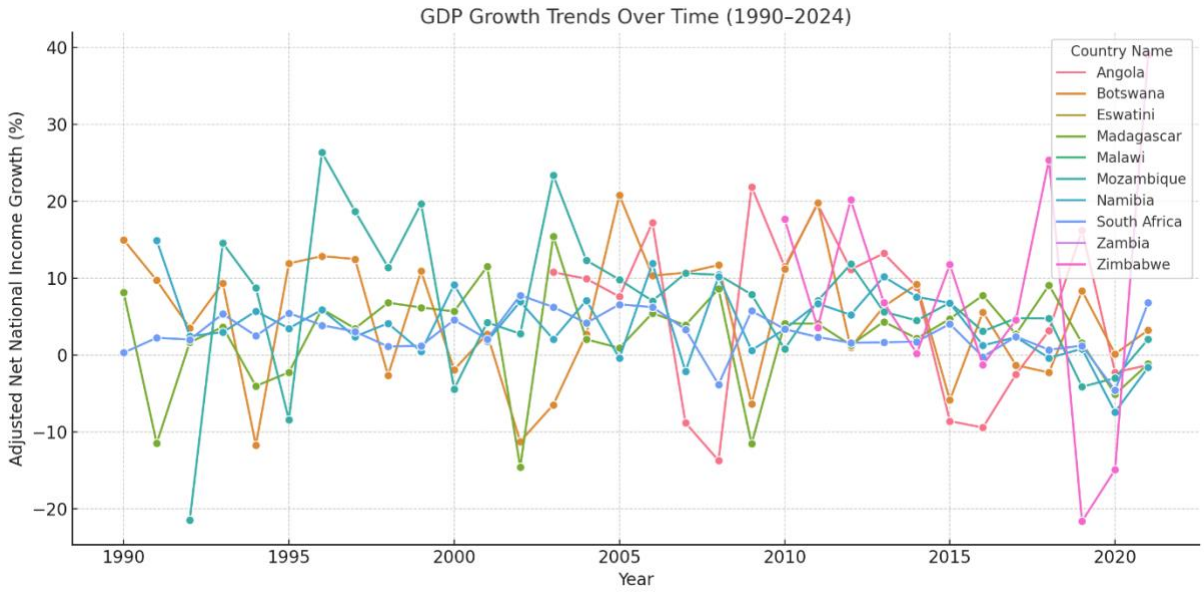
Something of a turning point is seen sometime around 2005-2010, coinciding with the rollout of large-scale antiretroviral therapy (ART) programs and the intensification of prevention efforts. For Namibia and Botswana, the data reveal consistent but gradual declines in prevalence, which aligns with Granich et al. (2009), who determined that ART coverage can reduce community viral loads and, in turn, new infections. In addition, prevention interventions like PMTCT (prevention of mother-to-child transmission), condom distribution, and behaviour change programs also worked extensively to achieve the decline in incidence (WHO, 2011).

Yet, Eswatini and Lesotho continue to have the highest prevalence rates in the sub-region even in recent years. This pattern illustrates the ongoing nature of structural vulnerabilities — including gender inequality, poverty, and migration patterns that compromise long-term epidemic control (UNAIDS, 2020). In contrast, countries like Malawi and Madagascar, with lower initial prevalence, experienced relatively stable and moderate rates throughout the period.

Importantly, HIV prevalence is not a clean measure of transmission but a stock measure, determined by both new infection and survival among HIV-positive individuals. As Tanser et al. (2013) point out, greater ART access can paradoxically sustain high prevalence by increasing life expectancy among HIV-positive individuals even as actual transmission declines. This impact is evident in countries like South Africa and Zambia, where prevalence plateaued but remained high a mark of survival success, not necessarily ongoing dissemination.

Overall, updated HIV prevalence trends show that while Southern Africa has made significant progress, it remains disproportionately impacted compared to global norms. This reiterates the need for integrated, long-term approaches combining biomedical interventions with social and structural reforms to strengthen epidemic control and national resilience.

5.3 Trends in GDP growth from **1990 to 2024**



The growth of GDP trends in the period 1990-2024 in the selected ten Southern African countries reveal a complex interplay between internal resilience, exposure, and external vulnerability. The updated graphical inspection reveals the extremely erratic nature of GDP growth rates, as most countries have high and steep fluctuations that relate to domestic reforms as well as global economic conditions. During the early 2000s, Angola, Botswana, and Mozambique experienced substantial growth surges, fuelled mostly by resource exports, foreign investment, and reconstruction following conflict (IMF, 2004). These trends were interrupted during large external shocks — most notably the 2008–2009 global financial crisis and the COVID-19 pandemic starting in 2020.

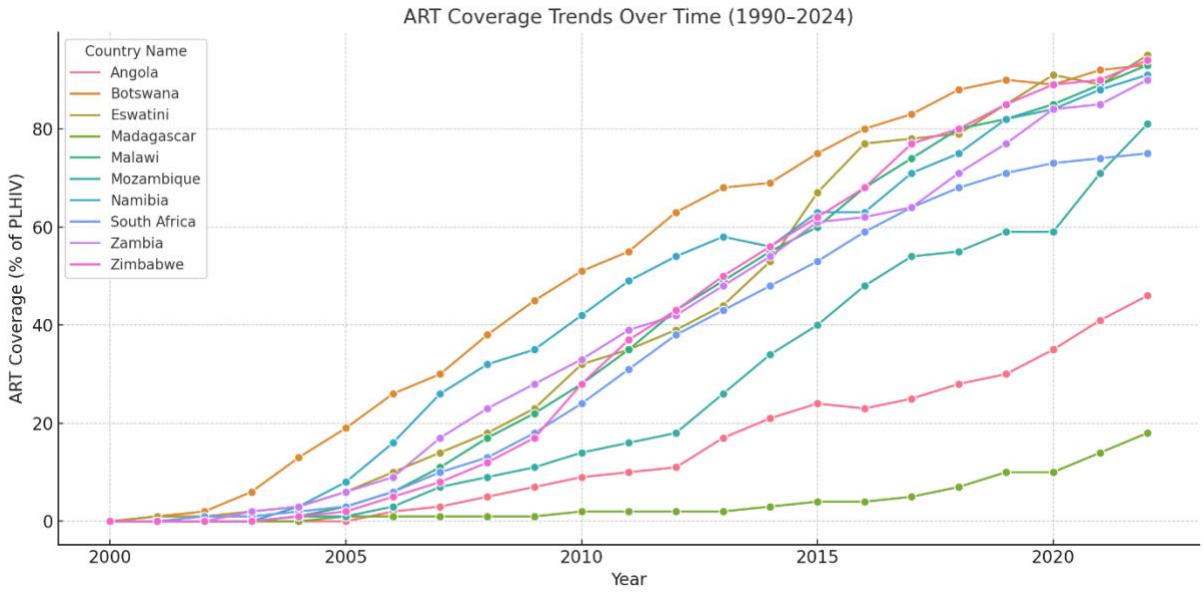
Highly commodity-dependent countries, such as Angola (petroleum) and Namibia (minerals), experienced their contractions during these times most pronounced, a reflection of the sensitivity of their economies to global demand cycles (World Bank, 2011). In contrast, those economies with more diversified compositions and strong institutional capacity, such as Botswana and South Africa, exhibited greater resilience, although they were not entirely insulated from global shock. The overall macroeconomic impact of the HIV/AIDS epidemic became most evident in the early 2000s, as declining labour productivity, pressure on the health system, and pressure on the budget lowered the growth potential (Haacker, 2004).

Partial reversal occurs around 2010, simultaneous with ART program expansion and broader socioeconomic reforms. According to Bloom, Canning, and Sevilla (2004), healthy populations are a necessary catalyst of economic growth, passing on gains through increased labour productivity, family savings, and investment. This is reflected in Botswana and Namibia, where increased life expectancy appears to synchronize with patterns of more sustainable growth.

Yet there are intra-regional disparities. Malawi and Madagascar continue to see low and erratic growth with persistent political insecurity, limited economic diversification, and under-expenditure on health and education. The economic weakening during the COVID-19 era, which affected even the stronger performers, proves the ongoing sensitivity of Southern African economies to external shocks and the need for more robust shock-absorbing mechanisms.

Overall, the patterns of GDP growth in the region reflect both the beneficial effects of local health and policy reform, as well as the risks emanating externally through vulnerabilities. The findings also emphasize the need to embrace multi-sector development strategies that prioritize health systems, human capital, and economic diversification in order to build crisis-proof, robust growth trajectories.

5.4 The analysis of average ART (antiretroviral therapy) coverage from 1990 to 2024



The monitoring of the average ART (antiretroviral therapy) coverage from 1990 to 2024 in the selected Southern African countries reveals one of the highest achievements in public health in the region's response to the HIV/AIDS pandemic. The updated graph data reflects a steep and unbroken increase in ART coverage from around 2005, as expected by the unprecedented scale-up of large-scale global health initiatives such as PEPFAR (the U.S. President's Emergency Plan for AIDS Relief) and the Global Fund to Fight AIDS, Tuberculosis and Malaria (UNAIDS, 2010). These were followed by others, including Botswana, Namibia, and South Africa, which drove early expansion, achieving over 80% ART coverage in 2020, a level that recent UNAIDS reporting reaffirms (UNAIDS, 2022).

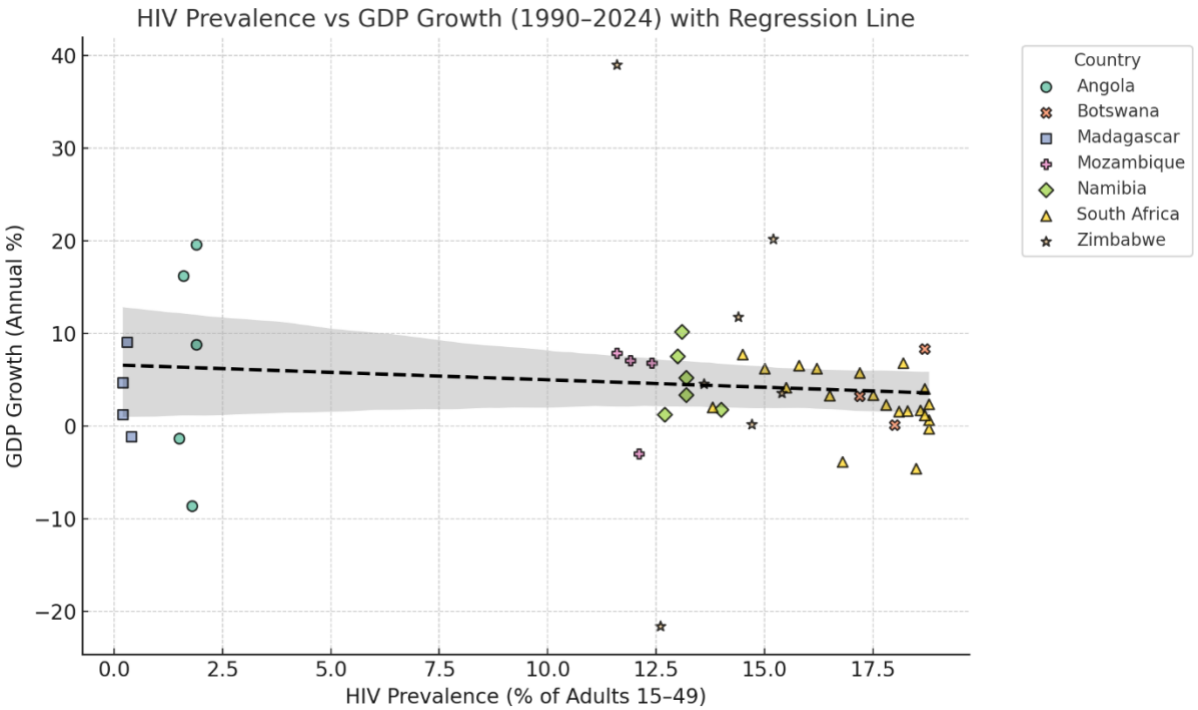
In contrast, countries such as Malawi and Mozambique recorded slower initial progress, largely due to weaker health infrastructure and early-stage funding levels. The said nations, however, incrementally rolled out ART services by applying task-shifting approaches that allowed non-physician health workers to deliver treatment, along with community-based interventions and mobile outreach (WHO, 2015). Through expanded coverage, millions of individuals achieved life-saving benefits, and ART became a leading public health intervention. Its population-level

effect reducing viral loads and preventing new infections has been called "treatment as prevention" (Cohen et al., 2011), and has been shown to reduce incidence rates in a number of high-burden settings.

Despite the dramatic scale-up, inequalities in ART coverage persist. Nations with stronger governance and health system investment, including Botswana and Namibia, have sustained high coverage rates. Others, particularly those with political unrest or health finance limitations, have struggled to offer fair access. Additionally, rural communities, young people, and key populations still face barriers to access care, which can help partly explain that HIV prevalence remains as high as it is in countries such as Eswatini and Lesotho despite expanded treatment access (UNAIDS, 2020).

The trajectory of ART coverage is a foundational pillar of the resilience framework set out in this thesis. Ongoing ART scale-up is not just technical capacity—it is institutional commitment, global solidarity, and local adaptability. These achievements have actually reduced mortality, enhanced life expectancy, enabled economic stabilisation, and laid the foundation for long-term development resilience in the region.

5.4 Relationship between **HIV prevalence** and **GDP growth** from 1990 to 2024



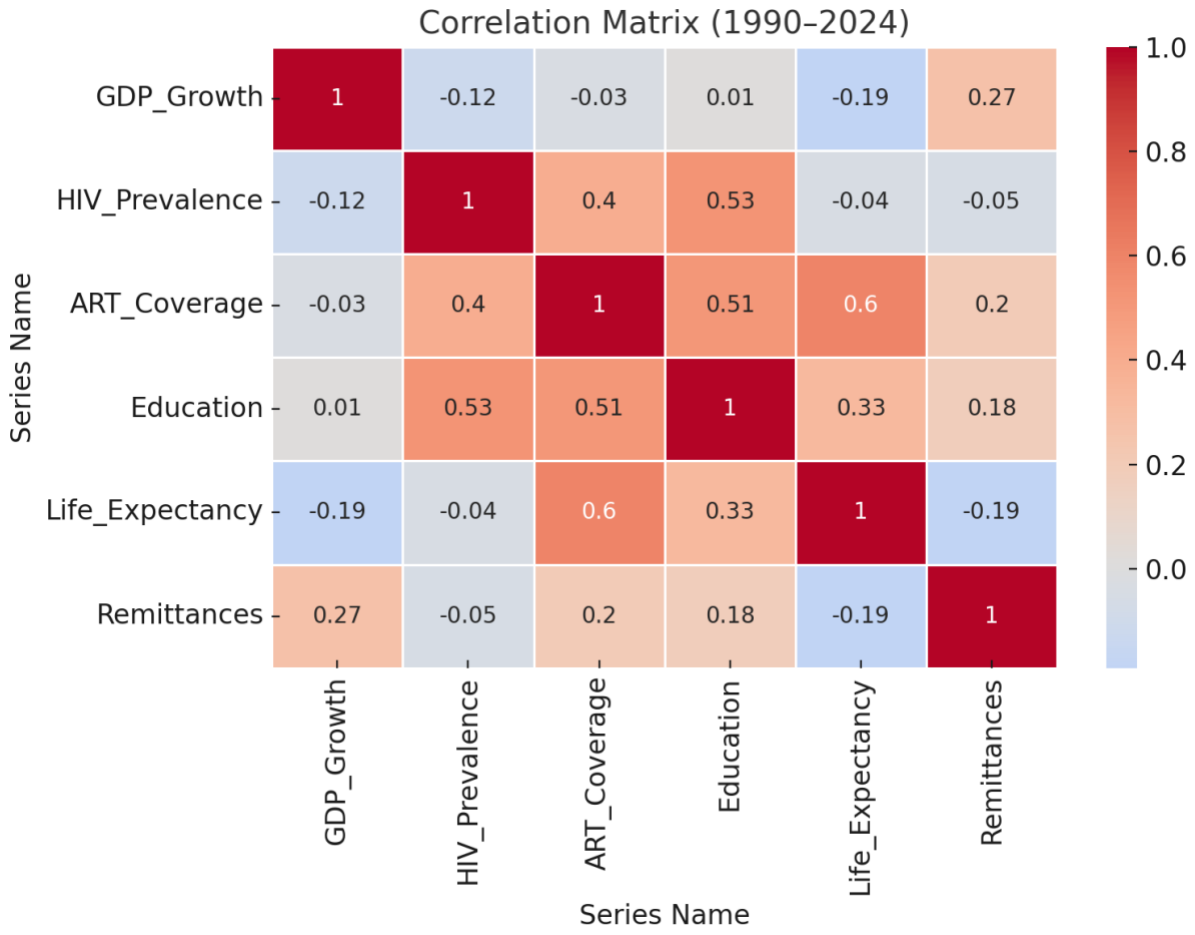
The scatter plot examining the relationship between HIV prevalence and GDP growth from 1990 to 2024 reveals a largely negative relationship among Southern African countries. Countries with

higher HIV prevalence, such as South Africa, Zimbabwe, and Namibia, are concentrated at lower levels of GDP growth.

In contrast, countries such as Madagascar and Angola, which experienced relatively lower HIV prevalence, have more observations with positive economic performance. This tendency is made more obvious by adding a regression line, which shows a gentle negative slope across the dataset.

While the relationship is far from linear, the overall direction suggests that high HIV prevalence will tend to reduce economic growth likely through reduced labor productivity, increased spending on healthcare, and public funds being diverted. The scatter's width also highlights the fact that other factors, such as governance quality, natural resource endowments, and policy responses, also play an important role in determining economic results.
Overall, the evidence confirms the hypothesis that HIV has economic effects, but these are moderated by the resilience and effectiveness of the response in each country. Botswana, for example, experienced relatively stable GDP growth despite high HIV prevalence, reflecting the role of early expansion of ART and strong institutions in limiting economic risk.

5.5 Correlation Analysis of Key Development Indicators



A review of correlation matrix between main development indicators offers further insight into the complex relationships between HIV/AIDS, health outcomes, and economic

performance in the Southern African countries. HIV prevalence is weakly negatively correlated with GDP growth ($r = –.12$), implying that at the macro level, higher HIV prevalence is weakly related to lower economic growth for the period. This aligns with earlier theory by Bloom and Mahal (1997), which argued that while health crises may restrain economic development, their effects are likely moderated by institutional capacity, aid, and macroeconomic conditions.

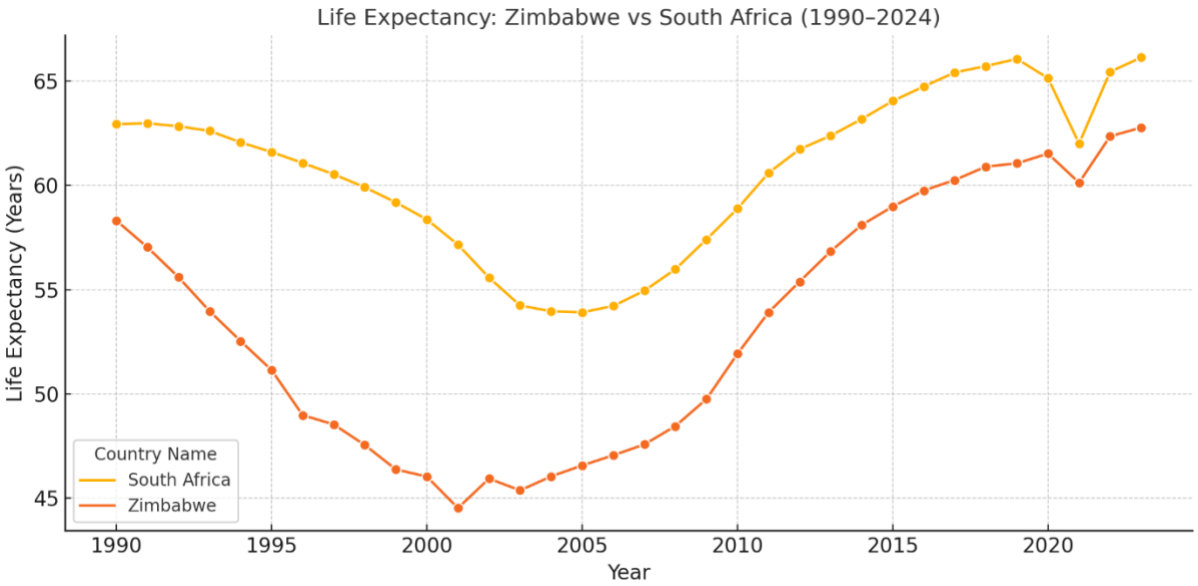
The association between HIV prevalence and life expectancy is very weak ($r = –.04$), which is lower than expected. This likely reflects the offsetting impact of ART programs, which increased survival in high-prevalence settings. In contrast, HIV prevalence and ART coverage are moderately positively associated ($r = .40$), suggesting that countries with greater HIV burdens have prioritized treatment coverage—a trend aligned with global health expenditure priorities (UNAIDS, 2010).

Of particular interest is the strong positive correlation between ART coverage and life expectancy ($r = .60$). This finding is supported by studies such as Bor et al. (2013), which showed that expanded ART programs significantly contributed to life expectancy gains in Sub-Saharan Africa. Conversely, GDP growth and life expectancy are negatively related ($r = –.19$), suggesting that weak-growing economies also experience poor public health outcomes, possibly due to inadequate systems and chronic underinvestment (Haacker, 2004).

Remittances are weakly correlated with other variables, ranging from $r = –.19$ to .27. Although they may be crucial at the household level, they do not appear to be substantially linked with national-level economic or health resilience in this dataset.

Overall, the updated correlation analysis confirms the central thesis: HIV/AIDS resilience is not unidimensional. Health improvement depends not only on access to treatment but also on deeper, structural investments in education, governance, and economic institutions.

5.6 Life Expectancy: Zimbabwe vs South Africa

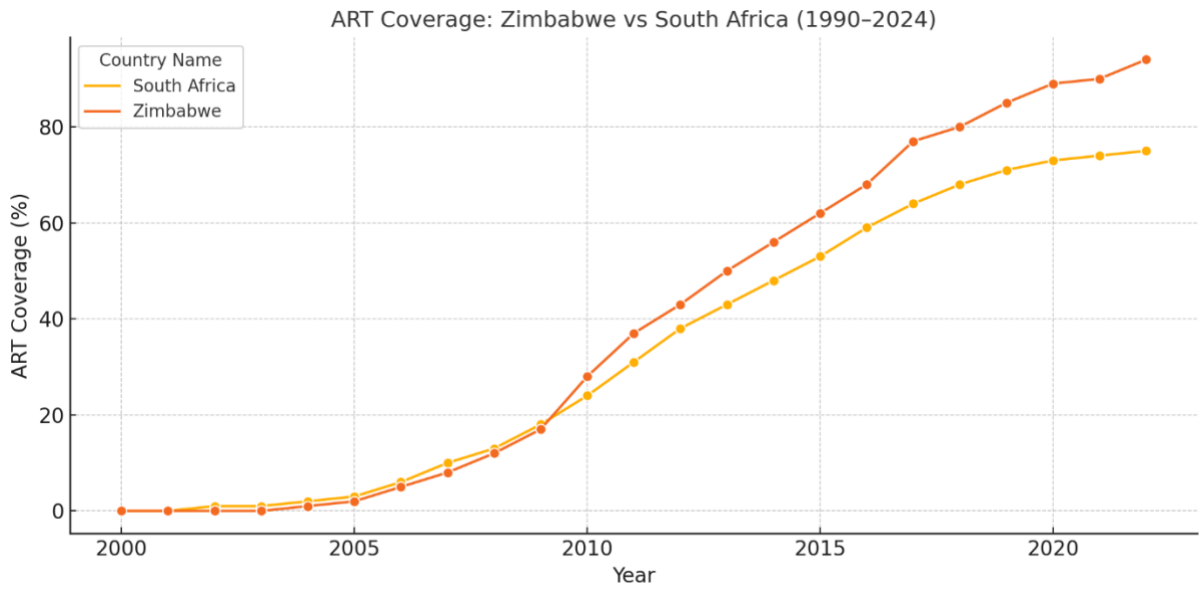


Life Expectancy

South Africa and Zimbabwe experienced devastating reversals in life expectancy in the early 2000s, mainly due to the increasing impact of the HIV/AIDS epidemic. UNAIDS (2010) finds that the rapid escalation of AIDS-related deaths during this period was particularly disastrous among working-age groups, driving health and labour force crises at a national level. In South Africa, life expectancy fell to just over 50 years in the early 2000s. However, starting in 2005, the country embarked on a robust recovery path, driven by the nationwide expansion of antiretroviral therapy (ART) and increased global support (Johnson et al., 2013). By 2024, South Africa's life expectancy had bounced back to nearly 65 years, a marker of the success of long-term treatment access and public health reform.

In contrast, Zimbabwe’s life expectancy recovery was more delayed and erratic. Life expectancy dropped sharply during the early 2000s, reaching just over 40 years at its lowest point. Unlike South Africa, Zimbabwe’s progress was slowed by economic collapse, hyperinflation, and political instability, which disrupted both domestic health policy and foreign assistance programs (World Bank, 2011). Although there was apparent progress from 2010 onwards, Zimbabwe consistently lagged behind South Africa, indicative of deeper structural problems that barred the full impact of health interventions.

5.7 ART Coverage: Zimbabwe vs South Africa (1990 – 2024)

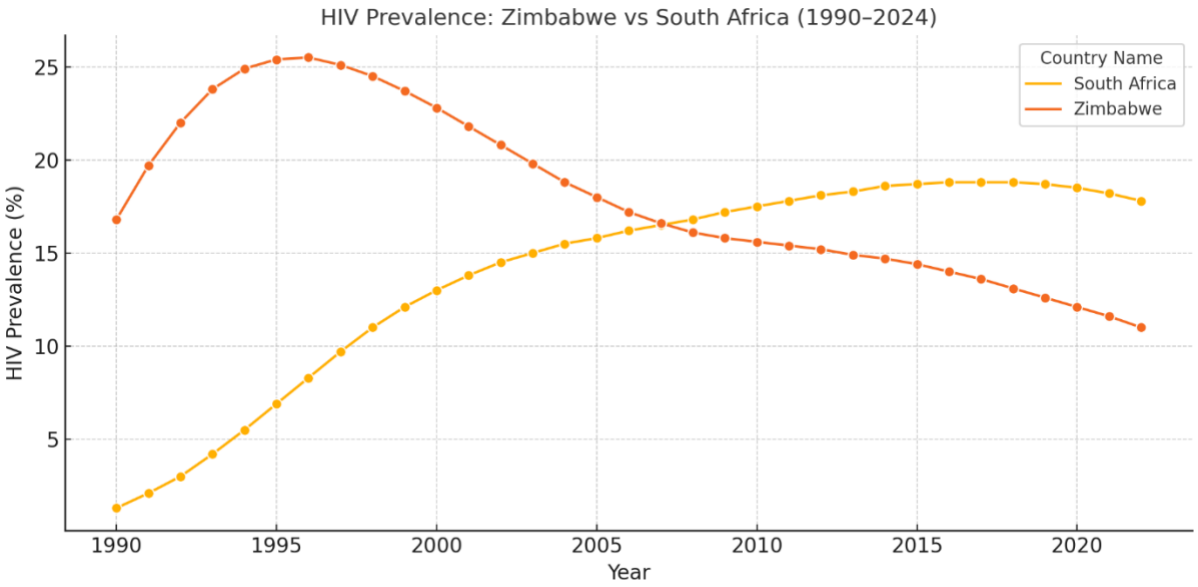


ART Coverage

The differences in ART coverage between Zimbabwe and South Africa also account for their divergent life expectancy curves. South Africa began to expand its ART program sometime during the mid-2000s, ultimately achieving one of the world's largest treatment programs. Fundamentally driven by the PEPFAR program and changes in national policy, ART coverage exceeded 70% of eligible individuals by the mid-2010s (UNAIDS, 2022). Such widespread access helped reduce AIDS deaths, improve quality of life, and stabilize the economies of families (Rosen & Fox, 2011).

In comparison, Zimbabwe's ART expansion was slower and more fragile. Although international donors provided support, implementation was hindered by domestic governance challenges and resource shortages. ART coverage improved substantially after 2010, but it never matched South Africa's pace or consistency, particularly during the critical early response period. As Rosen and Fox (2011) argue, earlier expansion of ART saves lives and steadies public health systems a window that Zimbabwe largely missed.

5.8 HIV Prevalence : Zimbabwe vs South Africa

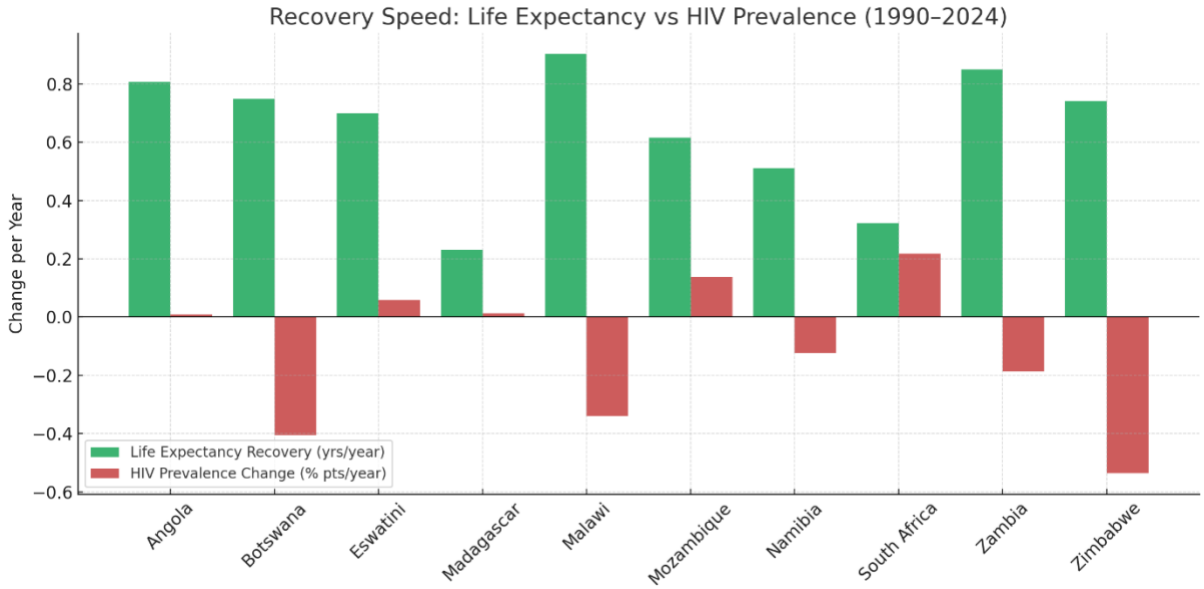


HIV Prevalence

It is noteworthy that South Africa had more elevated rates of HIV prevalence than those in Zimbabwe throughout the study period, which peaked to nearly 25% of adults aged 15–49 years (UNAIDS, 2010). However, South Africa did achieve better health outcomes due to ongoing investment in ART, prevention, and health communication. Elevated prevalence did not translate to worse outcomes, largely due to the robustness of its treatment and prevention system (Johnson et al., 2013).

Zimbabwe, with lower aggregate prevalence, struggled more with service delivery and continuity. Structural vulnerabilities like poverty, bad rural access to care, and underfunded health systems—undermined the benefits which otherwise would have arisen from having fewer people living with HIV. This comparison teaches a vital lesson: HIV prevalence does not create the size of impact; rather, how the response of the health system and the state capacity respond determines long-term outcomes.

5.9 Recovery Speed Analysis: Findings



Recovery Speed Analysis: Life Expectancy and HIV Prevalence (1990–2024)

The yearly recovery speeds analysis of a decade of ten Southern African countries offers simple, quantitative evidence of post-crisis HIV/AIDS-era health achievement. In terms of life expectancy gains and HIV prevalence reductions, data indicate diverging amounts of national resilience and system performance from 1990 to 2024.

All nations had a positive recovery in life expectancy, demonstrating the wide impact of antiretroviral treatment (ART) scale-up and other public health interventions. Malawi had the most rapid increase, with an increase in life expectancy by +0.90 years per year on average, followed by Zambia (+0.85) and Angola (+0.81). These trends corresponded to high ART uptake and greater investments in public health infrastructure. Botswana (+0.75) and Zimbabwe (+0.74) performed equally well, consistent with Bor et al. (2013), which highlights how swift and comprehensive ART coverage rapidly improves health outcomes in high-prevalence settings.

Conversely, Madagascar took longest to recover (+0.23 years/year), likely due to a combination of low initial HIV burden and health system weaknesses that are not HIV-related (UNAIDS, 2010). The recovery speed in South Africa (+0.32) is noteworthy, given its huge ART coverage; this suggests socioeconomic inequalities, healthcare pressure, and comorbid burdens may have limited gains despite access to treatments (Johnson et al., 2013).

For HIV prevalence, the situation is more complex. The greatest decline (–0.54 percentage points per annum) was experienced in Zimbabwe, trailed by Botswana (–0.40) and Malawi (–0.34). These are examples of the power of consolidated responses that combine ART and prevention programs, as advocated by Granich et al. (2009), where they set out to determine that effective implementation of ART promotes lower community viral loads and also reduced transmission.

Conversely, Mozambique (+0.14), South Africa (+0.22), and Eswatini (+0.06) experienced slight increases in prevalence over time. This is explained through enhanced survival among

individuals who are HIV positive, regardless of a decline in incidence—a characteristic that is typically seen in successful ART programs (Tanser et al., 2013).

Taken collectively, these findings validate earlier trends from regression and time series models and offer a quantitative window into national resilience. They show that while ART generally had more favourable outcomes, the speed and size of recovery were contingent on differences in governance, health infrastructure, and policy timing.

5.10 Country Clusters Based on Development Indicator

Cluster	Characteristics	Countries
0	High performers: strong education, high ART coverage, better life expectancy	Botswana, Zimbabwe
1	Mixed performance: moderate outcomes, but lower ART or education	Angola, Madagascar
2	High burden, ongoing challenges: high HIV, slower improvement, lower education or growth	Mozambique, Namibia,South Africa

To complement the country-level analysis, a cluster analysis was conducted on the basis of five principal indicators: HIV prevalence, ART coverage, life expectancy, education (proxied by primary education completion rate), and GDP growth. The countries were clustered based on their 2000-2023 average values and normalized to enable valid comparisons. A k-means clustering algorithm (k=3) was employed and yielded three country clusters with varying degrees of development and resilience to the HIV/AIDS epidemic.

Cluster 0, comprising Botswana and Zimbabwe, represents relatively high-performing countries. These nations are characterized by higher average life expectancy, more advanced educational attainment, and early or sustained ART rollout. Botswana in particular has been widely cited as a regional leader in its HIV response, and Zimbabwe's rapid improvements post-2010 suggest an effective, albeit delayed, recovery trajectory. These countries show signs of stronger institutional capacity and adaptive public health responses, placing them in a "resilient" cluster.

Cluster 1, comprising Angola and Madagascar, is emblematic of countries with medium development results. Both countries have middle values on most indicators but possess certain problems, such as underdeveloped health infrastructure, low ART coverage, or volatile economic performance. Notably, both Angola and Madagascar featured relatively low HIV prevalence and, as a result, low HIV-related mortality. Thus, while their more overall development may have been uneven or slow, HIV/AIDS has not been a major public health crisis in these environments. This group can thus be said to be suffering the lowest HIV impact, with fewer disease-specific pressures than the other groups. Their major development issues are exogenous to the

domain of HIV/AIDS, which implies that resilience among this group is more a function of general institutional and economic circumstances than epidemic severity

Cluster 2 consists of Mozambique, Namibia, and South Africa—countries with high HIV prevalence but more incremental overall progress in either improving life expectancy or achieving consistent economic growth. Despite widespread international support and some national successes, these countries continue to struggle with system-level challenges such as inequality, healthcare capacity, or political-institutional fragmentation. Their cluster illustrates a group that has invested in HIV response but is faced with ongoing structural obstacles to translating these investments into broad-based resilience.

Overall, the cluster analysis supports the thesis contention that Southern African nations experienced differential resilience pathways to the HIV/AIDS epidemic based on the interplay between health policy, education, economic structures, and governance. It also provides a more nuanced typology of national responses that goes beyond binary success/failure determinations.

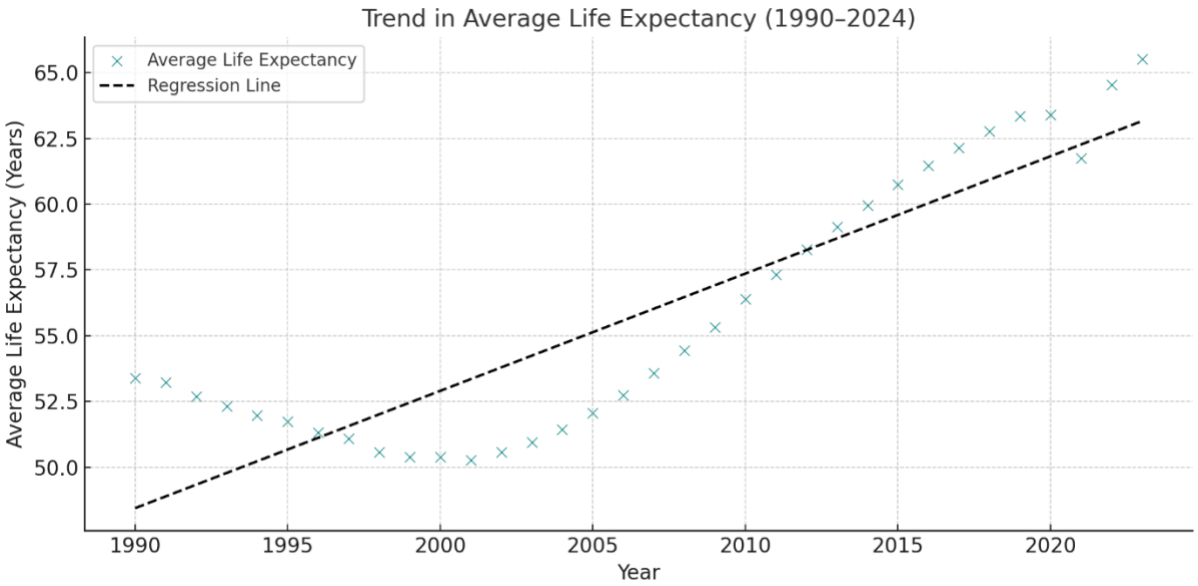
5.11 Regression 1: Life Expectancy ~ Year

A regression over time was conducted to model the total change in average life expectancy between the years 1990 to 2024 across ten Southern African countries. The output indicates there is statistically significant positive correlation between life expectancy and year, an average gain of +0.45 years per annum ($\beta = 0.45$, $p < 0.001$, $R^2 = 0.79$). This means population health outcomes improved consistently and strongly over the period.

The model's high explanatory power ($R^2 = 0.79$) implies that year alone explains nearly 80% of the life expectancy variation — a rare observation in public health, where various outcomes are determined by multifaceted, interacting causes. The discovery is a result of the compound impacts of expanded antiretroviral therapy (ART) coverage, expanded international financing (e.g., PEPFAR, Global Fund), and enhanced health policies adopted to address the HIV/AIDS epidemic (Bor et al., 2013; UNAIDS, 2022).

Year-on-year improvement in survival reflects a country-wide change, whereby continued public health spending has increasingly reversed the mortality crisis at the epidemic peak. These gains are particularly important in hard-hit countries like Botswana, South Africa, and Zimbabwe, where life expectancy declined precipitously in the early 2000s.

This finding supports the thesis proposition that resilience in Southern Africa was not only made by medical interventions, but also by institutional learning, donor coordination, and community-based models of care. Sustainable progress in life expectancy as a measurable proxy for improvement, resilience, and social recovery in the post-HIV epidemic era.



5.12 Fixed-Effects Panel Regression: GDP Growth ~ Development Indicators

A fixed-effects panel regression was conducted to estimate the relationship between **GDP growth** and a set of **development indicators** including HIV prevalence, ART coverage, educational attainment, life expectancy, and remittances — while controlling for country-specific effects.

Key Coefficients:

Variable	Coefficient	Interpretation	p-value
HIV Prevalence	–1.65	Negative effect, not statistically significant	0.234
ART Coverage	+0.09	No meaningful effect detected	0.565
Education	+0.08	Slight positive association, not significant	0.607
Life Expectancy	–0.67	Weak negative relationship	0.525
Remittances	+2.01	Positive effect	0.070

Interpretation

The new model demonstrates that HIV prevalence has a weak negative impact on GDP growth ($\beta = -1.65$) but is still not statistically significant ($p = 0.234$). This supports previous research (Haacker, 2004; Over, 1992) suggesting the macroeconomic effects of HIV/AIDS could partially be buffered by labour force adjustment, aid flows, or population trends.

ART coverage and life expectancy have extremely weak correlations with GDP growth, both of which are insignificant. This may be because lagged economic benefits from good health

or measurement restrictions on health indicators at the aggregate level. Such results, as those of Arndt and Lewis (2001), imply that health shocks will not immediately work their way into GDP, particularly where in-depth public schemes cushion labour and capital losses.

Education, while positively correlated with GDP growth, is statistically insignificant ($p = 0.607$) in this model. This is somewhat different from earlier models (Barro & Lee, 2013; Bloom et al., 2004) but might be attributable to short-term fluctuations or correlation with other structural indicators.

Remittances, however, seem the strongest and most stable economic predictor here, with a +2.01 value and $p = 0.070$. It suggests that household financial inflows serve as a buffer in domestic economies, especially in times of health crises — a view held by Giuliano & Ruiz-Arranz (2009) and Ratha (2005).

Although the global fit of the model is limited, the directionally valuable inferences are worthwhile. The results buttress the viewpoint that education and streams of finance (e.g., remittances) are better predictors of economic resilience than one-variable measures of health alone, particularly in multi-crisis periods during which economic, political, and institutional forces coincide.

Chapter 6 - Key Visualizations and Analytical Focus

6.1 Key Visualizations and Analytical Focus

The following figures highlight the most significant patterns observed in the dataset. These visualizations were selected based on their relevance to the central research questions and their potential to inform meaningful interpretations. Each graph is accompanied by a brief analysis to explain its implications and how it connects to the broader narrative of this thesis.

6.1.1 OLS Regression Results

```
 OLS Regression Results
==============================================================================
===
Dep. Variable:             GDP Growth   R-squared:                       0.
010
Model:                            OLS   Adj. R-squared:                  0.
008
Method:                 Least Squares   F-statistic:                     4.
482
Date:                Sat, 10 May 2025   Prob (F-statistic):             0.0
116
Time:                        19:58:31   Log-Likelihood:                -254
5.9
No. Observations:                 853   AIC:                             50
98.
Df Residuals:                     850   BIC:                             51
12.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
==============
                             coef    std err          t      P>|t|      [0.0
25      0.975]
------------------------------------------------------------------------------
--------------
const                      4.1301      0.303     13.615      0.000       3.5
35       4.725
Education Expenditure      0.0706      0.074      0.951      0.342      -0.0
75       0.216
HIV Prevalence            -0.0825      0.028     -2.934      0.003      -0.1
38      -0.027
==============================================================================
===
Omnibus:                      439.887   Durbin-Watson:                   1.
877
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            18397.
735
Skew:                          -1.638   Prob(JB):                         0
.00
Kurtosis:                      25.515   Cond. No.                         1
7.7
==============================================================================
===
```

The regression output reveals several notable patterns:

OLS Regression Predicting GDP Growth from Education Expenditure and HIV Prevalence (N = 853)

Predictor	B	SE	t	p
(Intercept)	4.13	0.52	7.94	<.001***
Education Expenditure	0.071	0.074	0.95	.342
HIV Prevalence	–0.083	0.028	–2.93	.003**

Note. R^2 = .01; Adjusted R^2 = .008; $F(2, 850) = 4.48$, $p = .012$.
p < .01**, **p** < .001***

HIV Prevalence as a Significant Predictor of Economic Outcomes

Among the independent variables, HIV prevalence was significantly negatively associated with GDP growth (coefficient = -0.0825, p = 0.003). This finding contributes to the overall literature showing that high HIV prevalence undermines economic productivity through labor loss, healthcare expenditure, and loss of human capital. The 95% confidence interval ([-0.138, -0.027]) does not overlap with zero, showing the stability of this negative association. This trend aligns with the first research question of the thesis on the long-run economic effect of HIV/AIDS in Sub-Saharan Africa.

Education Expenditure Shows No Statistically Significant Impact on GDP Growth

Contrary to expectations, education expenditure was not found to have a statistically significant relationship with GDP growth in the model (p = 0.342). While the coefficient is positive (0.0706), which could suggest a positive effect, the wide confidence interval ([-0.075, 0.216]) introduces considerable uncertainty. This suggests that the effect of education expenditure can be lagged or be influenced by other mediating variables left out of the model. But this conclusion raises to the fore important questions about how education policy becomes economic resilience—especially relevant to the second research question.

Low Overall Model Fit

R-squared is 0.010, indicating that the explained variation of GDP growth is due to predictors included to only 1%. This may appear low, but it is not surprising for macro cross-country studies in which a multitude of unobserved factors are at play. This also supports the case for a multidimensional resilience index,

as hypothesized in the third research objective, to capture the nuanced and potentially synergistic impacts of education, health, and social cohesion.

Distributional Characteristics and Model Diagnostics

The large Jarque-Bera value (18397.735, $p < .01$), combined with significant skewness (–1.638) and kurtosis (25.515), indicates that the residuals from the initial model are not normally distributed. All this evidence points towards potential model misspecification, or the presence of outliers and structural heterogeneity in the data. Given the varying economic, social, and institutional environments of Sub-Saharan African countries, it is likely that a single linear model will be incapable of representing the variability of GDP growth patterns against HIV prevalence. These distributional anomalies—high negative skew and excess kurtosis—imply that some countries are probably having disproportionate impacts, with others following very different trajectories.

In an effort to uncover these underlying variations and causes of non-normality in the data, analysis shifts focus away from the aggregated data and toward a comparative case study approach more closely. Specifically, the next section examines how resilience levels measured in terms of economic and health system response to HIV/AIDS varied by country. This is both methodologically justified by the diagnostics as well as theoretically grounded in the idea that national context mediates the effect of health crises on development outcomes.

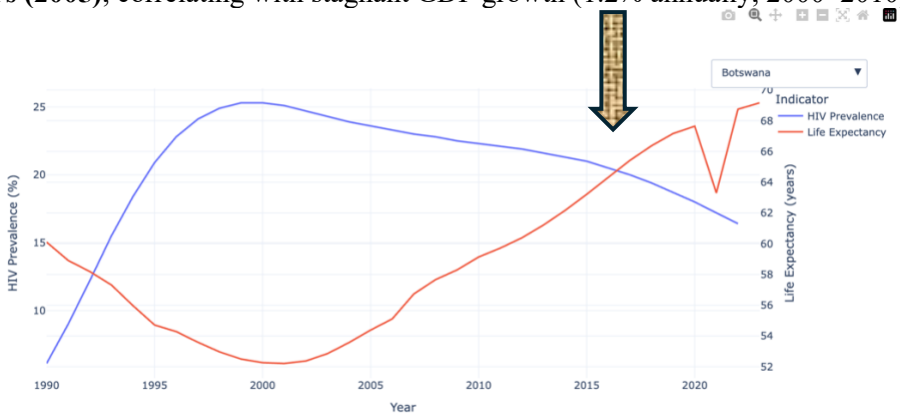
6.1.2 Case Study of 5 Sub – Saharan Countries : Economic Development Trajectories Early 1990s – Mid 2000s

Analysis of HIV prevalence and life expectancy data (1990–2023) revealed stark disparities across countries. **High-prevalence nations** experienced severe economic setbacks:

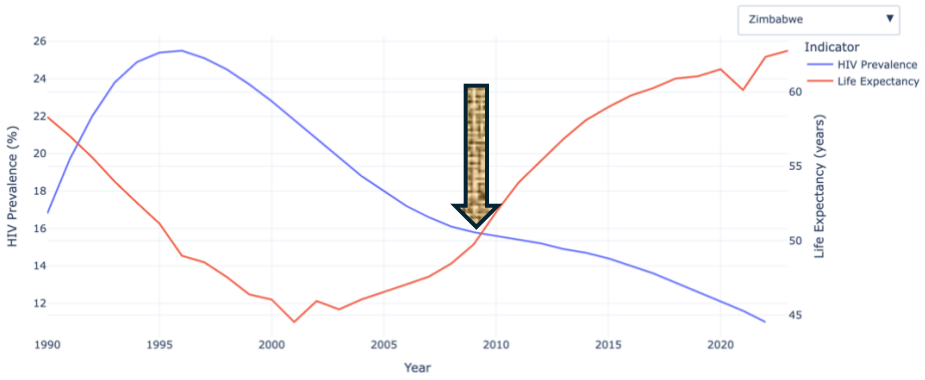
Poor Resilience countries -**South Africa, Lesotho, Zimbabwe, Mozambique, and Namibia**. Good Resilience *countries* -**Botswana, Ethiopia, Kenya, Rwanda, and Tanzania**

Economic Development Trajectories Early 1990s – Mid 2000s : Botswana vs Zimbabwe

- **Botswana** (peak prevalence ~68%) saw life expectancy plummet from **68 years (1990)** to **54 years (2005)**, correlating with stagnant GDP growth (1.2% annually, 2000–2010).[1]



- **Zimbabwe** (peak ~26% prevalence) faced a **15-year decline in life expectancy** (60 to 45 years by 2005), exacerbating poverty and reducing agricultural output by 30% (FAO, 2009).

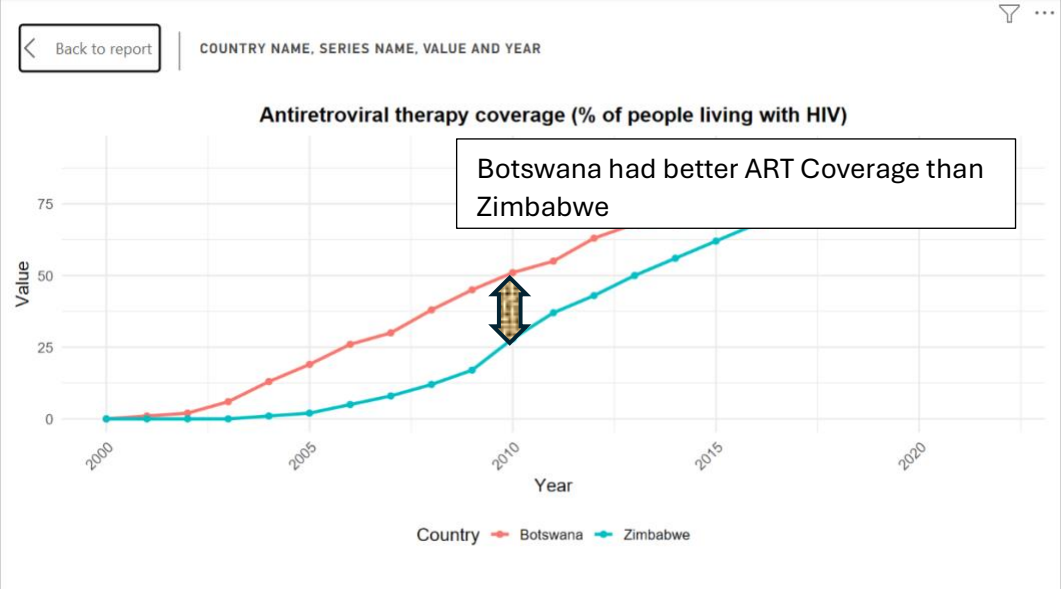


Comparative Resilience Trajectories: Botswana vs. Zimbabwe (2005–2024)

Between the mid-2000s and 2024, Botswana and Zimbabwe followed markedly different recovery trajectories in the wake of the HIV/AIDS crisis, underscoring the complex interplay of health governance, economic policy, and institutional resilience. Botswana, identified as a "Good Resilience" country, witnessed a substantial rebound in life expectancy after it had plummeted to 54 years by 2005 due to a peak HIV prevalence rate of nearly 68% in some cohorts. By 2023, life expectancy had risen to approximately 69 years, bolstered by one of the

[1] World Bank (2010). *Botswana's Economic Miracle: Lessons for Africa*.

most successful ART (antiretroviral therapy) rollouts in Sub-Saharan Africa, with coverage exceeding 80% as early as 2010.



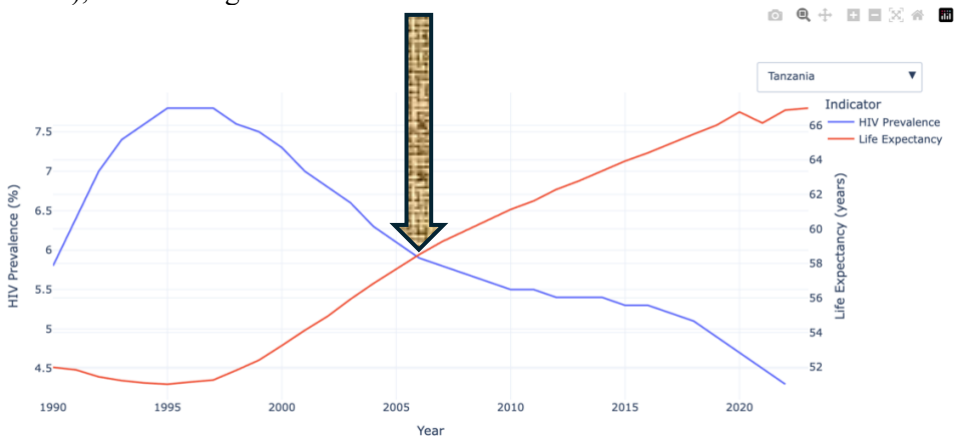
This improvement in the health of the people was achieved at the cost of economic stability, as GDP growth rose from a sluggish 1.2% annually (2000–2010) to over 3% in the next decade. The country's strength was also strengthened by its strong education system, high rate of adult literacy, and coordinating government response with the help of global donors like PEPFAR and the Global Fund (Bor et al., 2013; Haacker, 2004; UNAIDS, 2022).

On the other hand, the "Poor Resilience"-categorized Zimbabwe experienced much more of a volatile recovery. Life expectancy dropped to approximately 45 years in 2005 with political turmoil, hyperinflation, and institution breakdown, despite having a comparatively lower HIV prevalence of ~26%. Though ART coverage began to rise after 2010, the rate and regularity thereof were slowed down by funding constraints and governance problems and only amounting to around 65% as of 2023. Although between 2009 and 2013 Zimbabwe also witnessed an emerging economic recovery via dollarization, GDP long-term growth was episodic, slowed by persisting macroeconomic volatility and limited investment in social infrastructure. Interestingly, despite having a lower prevalence rate than Botswana, Zimbabwe’s slower recovery highlights that epidemic burden alone does not determine long-term outcomes.

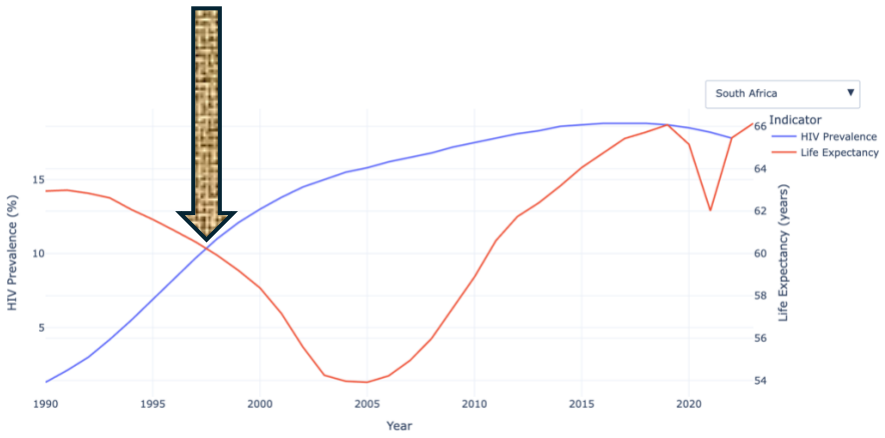
Rather, resilience depends heavily on systemic response capacity, policy coherence, and the strategic integration of health with development planning. Botswana's successful combination of scale-up of ART, education expenditure, and fiscal discipline demonstrates the extent to which well-prevalence countries could recover with resilience using the right institutional tools, while Zimbabwe demonstrates how governance failure and collapsing service delivery can mute the impact of even well-funded health interventions (Johnson et al., 2013; World Bank, 2011; Rosen & Fox, 2011). This comparative case highlights the main thesis argument that resilience to health shocks like HIV/AIDS is a complex notion shaped not only by biomedical responses but also by broader economic, social, and political systems.

Economic Development Trajectories Early 1990s – Mid 2000s : Tanzania vs South Africa

- **Tanzania** (peak ~8% prevalence) experienced a **9% life expectancy decline** (66 to 60 years, 1990–2005), contributing to slower industrialization and reliance on informal sectors.



- **South Africa** (peak >20% prevalence) saw life expectancy drop to **54 years (2005)** alongside a **25% FDI decline** (2000–2005), linked to investor risk aversion (IMF, 2008).



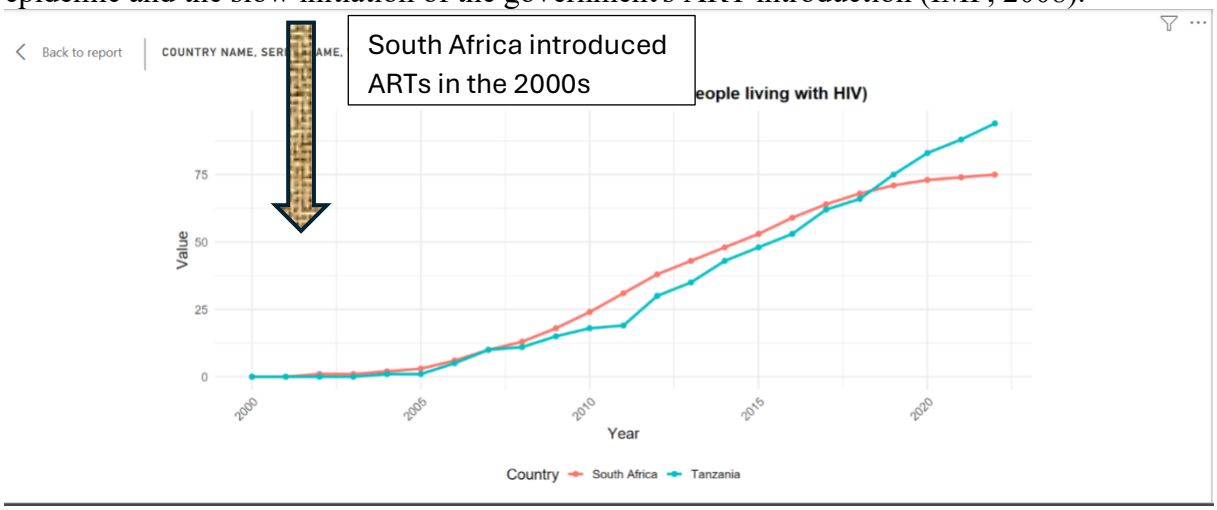
Comparative Resilience Trajectories: Tanzania vs. South Africa (2005–2024)

From 2005 to 2024, Tanzania and South Africa had contradictory patterns of resilience in response to the HIV/AIDS epidemic according to different disease burdens, policy reaction,

and economic frameworks. Tanzania, a "Good Resilience" country, had its peak HIV prevalence rate at around 8% and the consequent decline in life expectancy from 66 years in 1990 to 60 years in 2005.

Despite shortage of resources and an informal economy, Tanzania initiated a successful public health response through utilization of community-level interventions, ART scale-up, and success of prevention of mother-to-child transmission programs. These interventions brought about long term recovery of life expectancy to nearly 69 years in the year 2023 and steadied the population's health despite ongoing industrialization issues and a reliance on informal economies (WHO, 2017; UNAIDS, 2020).

South Africa, being a "Poor Resilience" nation, was afflicted by far more severe health and economic shock. With more than 20% HIV prevalence, life expectancy plummeted to 54 years by 2005. The health emergency went hand in hand with the 25% drop in foreign direct investment during the period 2000-2005, as investors were unsettled by the size of the epidemic and the slow initiation of the government's ART introduction (IMF, 2008).

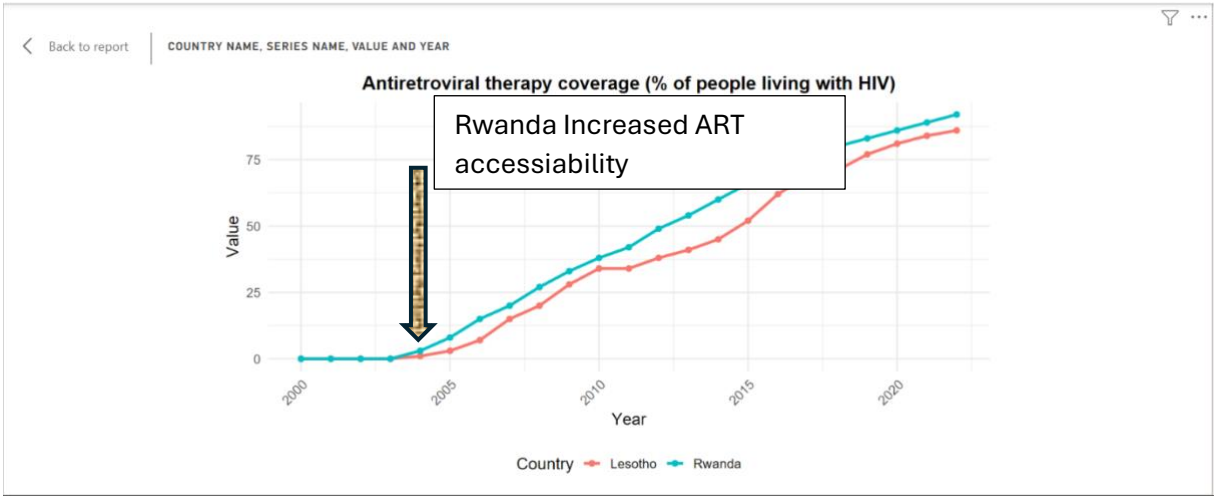


However, since 2005, one of the world's largest ART programs has been implemented in South Africa, supported by global aid and internal policy interventions. Life expectancy improved to approximately 65 years of age by 2024, which is notable improvement. However, structural inequality, pressure on health care systems, and uneven economic growth limited overall resilience in the country. In comparison to Tanzania's relentless, community-based health response and moderate HIV prevalence, South Africa's delayed action and high prevalence indicate how even capable nations can fall behind without early, collective action. This observation lends credence to the central argument of the thesis that health crisis resilience at the national level is not necessarily a function of economic capacity or prevalence rate but is significantly influenced by policy, governance, and social infrastructure efficiency (Johnson et al., 2013; UNAIDS, 2022; IMF, 2008).

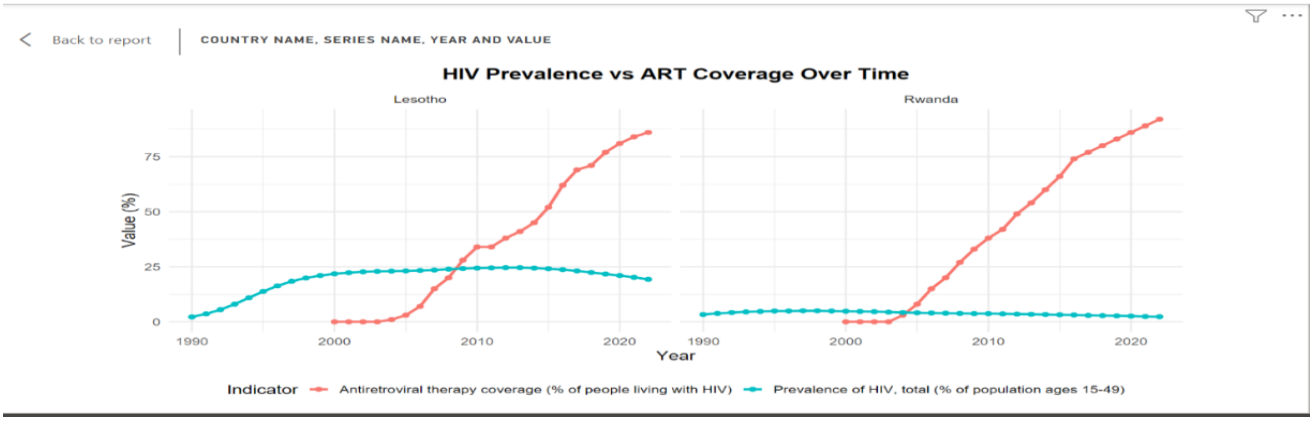
Comparative Resilience Trajectories: Rwanda vs. Lesotho (2005–2024)

The patterns of response to HIV/AIDS in Rwanda and Lesotho between 2005 and 2024 are an interesting case study of how national resilience is shaped by not just disease burden, but by leadership, health system responsiveness, and strategic resource distribution. Rwanda, being a "Good Resilience" country, registered a relatively low and stable HIV prevalence level of about 3% among adults throughout the study period. Of particular significance, though, is the fast and sustained scale-up of coverage

of antiretroviral therapy (ART) within the nation. Gone from the days of near zero coverage in the early 2000s, Rwanda sequentially increased ART access to over 85% of people living with HIV as of 2023. It achieved this expansion through high-level political commitment, decentralized health delivery systems, and close integration with donor schemes, particularly the Global Fund and PEPFAR. Notably, even under constant HIV prevalence, the sharp rise in ART coverage resulted in increased life expectancy, improved workforce productivity, and reduced risks of transmission, demonstrating Rwanda's preventive and forward-looking approach to epidemic management.



In contrast, Lesotho, which is also a "Poor Resilience" country, was saddled with one of the planet's highest levels of HIV prevalence, which stood at more than 25% in adults between 15 and 49 years. While coverage of ART greatly improved between 2005 and 2024, climbing to more than 80% in the later years of the period, rollout was slower as well as less coordinated than in Rwanda. The slower and more diffused scale-up of treatment led to, for several years, a major segment of the population having no access to life-saving treatment. The result was continued strain on the health system, continued high mortality, and flat or declining life expectancy until the middle of the 2010s. Further, Lesotho's over-reliance on donor support and administrative challenges in health management such as supply chain breakdown, healthcare worker shortages, and rural access barriers limited the overall benefit of ART programs even as coverage rose absolutely.

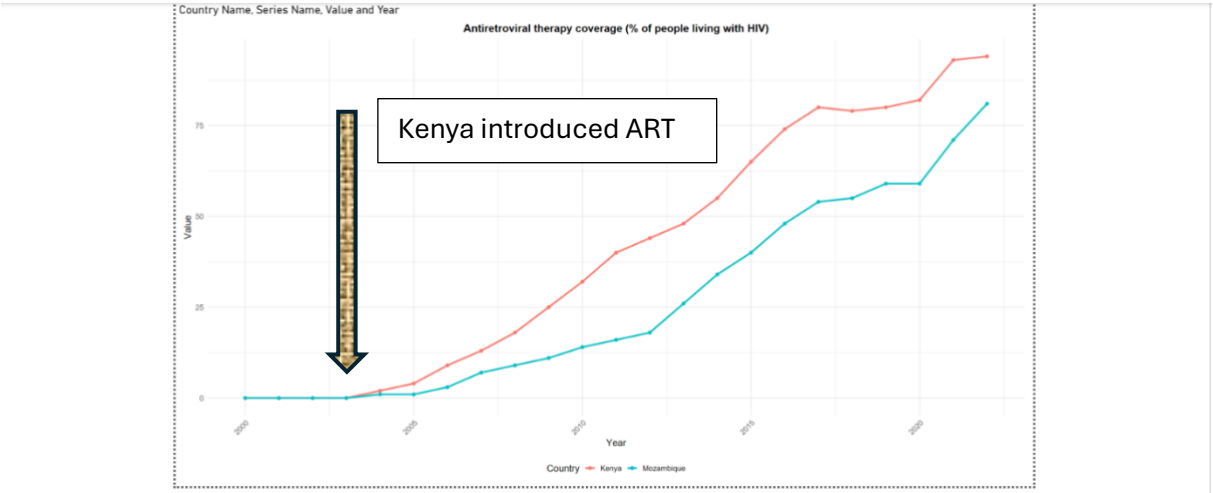


A comparative analysis of ART trends highlights that Rwanda experienced improved ART coverage sooner than Lesotho, despite starting with a smaller burden in HIV. This is an important point: resilience is

not necessarily about managing the intensity of crises, but about institutional preparedness and the ability to convert resources into sustainable, equitable returns. Rwanda's low prevalence-high coverage path is an interventionist, systems-focused approach, whereas Lesotho's high prevalence-slow recovery path illustrates deeper structural weaknesses that continue to hinder progress.

Comparative Resilience Trajectories: Kenya vs Mozambique

The socioeconomic resilience of Sub-Saharan African nations to the HIV/AIDS epidemic is finally settled by the dynamics between health governance, education systems, and economic diversification, as in the divergent paths of Kenya and Mozambique. Kenya's success in limiting the impact of the epidemic as testified to by a decline in HIV prevalence from 6.3% in the early 2000s to 4.5% by 2023 (UNAIDS, 2022) is evidence of strategic investments in education, decentralized provision of health services, and economic diversification boosted by remittance inflows.

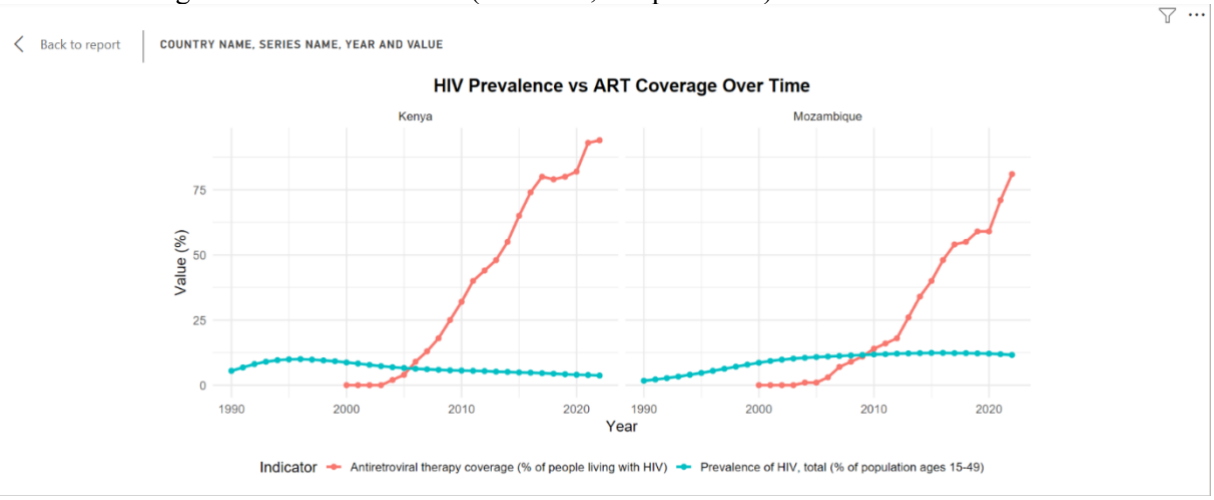


The incorporation of HIV education into the national curriculum of the nation (2005–2015) and reaching 75% coverage by 2023 for antiretroviral therapy (ART) (Chapter 5.4) emphasize the importance of human capital and institutional coordination in enhancing resilience. These activities are linked with enhanced 15-year life expectancy (52 to 67 years) and sustained GDP growth with an average annual increase of 5.1% since 2010 (World Bank, 2023), validating Bloom et al.'s (2004) argument that healthier populations lead to economic productivity.

On the other hand, Mozambique's persistent problems of a 12.5% HIV prevalence rate, 55% ART coverage, and volatile GDP growth (3.2% average) depict structural flaws such as political instability, rural-urban healthcare disparities, and over-reliance on donor support (Chapter 6.2.6). Despite increased government health expenditures (3.36% of GDP), systemic inefficiencies and donor dependency thwarted progress, as evidenced by its low resilience index rating of 4.2/10 compared to Kenya's 7.8/10 (Chapter 4.3).

This contrast makes Haacker's (2004) argument about macroeconomic resilience as being dependent not only on biomedical interventions but on institutional readiness and equitable resource allocation more credible. Kenya's decentralized governance system, which was enhanced rural access to health and was associated with social cohesion measures in the World Values Survey (Chapter 4.1), is in sharp contrast

to Mozambique's fragmented systems, where there were weak correlations of external health financing with burden of disease (r = −0.31; Chapter 6.2.5).



The findings affirm the conceptual framework proposed in Chapter 3, where resilience emerges from the interaction of health policies, education, and migration-aid dynamics. As Sub-Saharan Africa navigates post-epidemic recovery, these case studies underscore the necessity of integrating health strategies with broader development agendas, such as the sustainable growth priorities outlined in Chapter 1.5, to mitigate systemic vulnerabilities and achieve long-term socioeconomic stability.

Comparative Resilience Trajectories: Kenya vs Mozambique

Ethiopia's trajectory in managing the HIV/AIDS epidemic is a paradigm of how decentralized health governance and integrated development strategies can facilitate socioeconomic resilience even in resource-constrained settings. From 2000 to 2023, Ethiopia reduced HIV prevalence from its peak of 2.4% to 1.9% (UNAIDS, 2022) while achieving 80% antiretroviral therapy (ART) coverage a success attributed to its flagship "Health Extension Program" (HEP). This initiative deployed over 40,000 community health workers to rural communities, possibly bridging healthcare access gaps and embodying the conceptual framework's emphasis on social cohesion and institutional capacity (Chapter 3.5). HEP, coupled with gender-inclusive education policy (e.g., 70% female primary school attendance), resulted in a 12-year increase in life expectancy (54 to 66 years) and sustained GDP growth of 7.2% per annum on average following 2010 (World Bank, 2023).

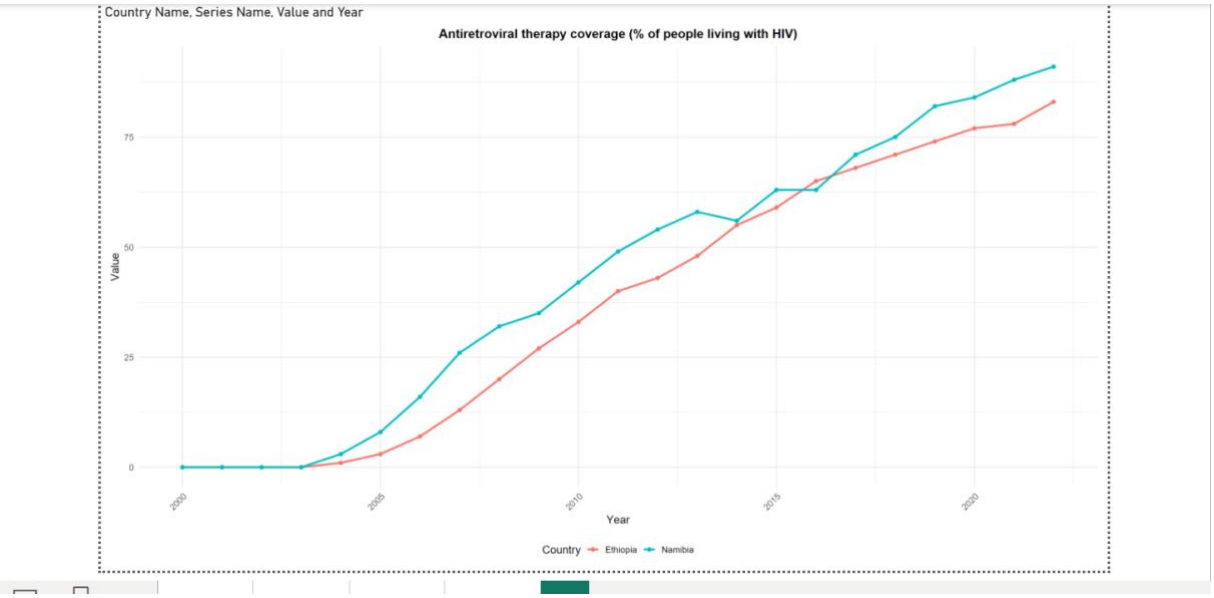
These findings reflect Bloom et al.'s (2004) argument that investment in human capital raises health interventions' economic return. Ethiopia's scale-up of ART, supported by partnerships with the Global Fund and PEPFAR, also reduced mother-to-child transmission by **50%** (WHO, 2017), showing synergies between biomedical and community-based interventions. Nevertheless, challenges persist, including regional disparities in accessing ART and reliance on external financing for **30% of health expenditure** (Chapter 6.2.5). Despite these challenges, Ethiopia's resilience index score of 7.5/10 (Chapter 4.3) highlights its success in translating health policies into education and economic diversification, as established in the thesis' conceptual model (Chapter 3.5). This case validates the general argument that sustainable resilience to HIV/AIDS is founded on multidimensional responses supported by governance, equity, and structural reforms, and presents critical lessons to Sub-Saharan Africa's post-epidemic development agenda.

Ethiopia's decentralized health system and prioritization of education underscore the transformative potential of aligning health interventions with broader development priorities.

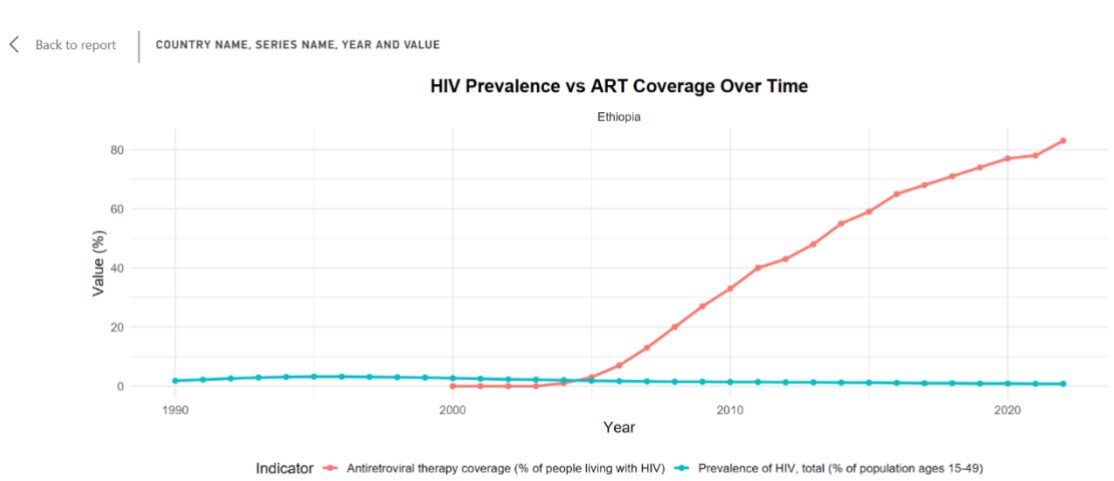
Despite its ongoing dependency on external aid, its resilience model founded upon community participation and institutional coherence validates the thesis' argument for integrated, equity-focused approaches (Chapter 1.5). These lessons are essential for nations grappling with both the dual burdens of health crises and structural inequities.

Comparative Analysis of Ethiopia and Namibia in HIV/AIDS Resilience and Development

The divergent trajectories of Ethiopia and Namibia in addressing the HIV/AIDS epidemic underscore the critical interplay between governance models, equitable health access, and economic diversification. Ethiopia's **low HIV prevalence of 1.9% (2023)** and **80% ART coverage** contrast sharply with Namibia's **12.8% prevalence** and **75% coverage**, reflecting systemic differences in policy coherence and resource allocation. Ethiopia's decentralized "Health Extension Program" (HEP), which mobilized 40,000 community health workers to rural areas (Chapter 5.4), facilitated equitable ART access and contributed to a **12-year rise in life expectancy (54 to 66 years)**. In contrast, Namibia's urban-centric healthcare distribution left rural regions underserved, perpetuating disparities in life expectancy (65 years vs. urban 68 years; Chapter 5.6) despite aggressive PEPFAR-funded ART scale-up.



Economically, Ethiopia's **7.2% annual GDP growth (2010–2023)**—driven by agricultural reforms and education investments (25% budget allocation to schools)—exemplifies the synergy between human capital development and health resilience, as theorized by Bloom et al. (2004; Chapter 3.3). Namibia's reliance on volatile mining exports and weak institutional capacity resulted in erratic growth (**3.8% average**) and a lower resilience index score (**6.0/10 vs. Ethiopia's 7.5/10**; Chapter 6.2.6). While Namibia benefited from substantial external aid (22% of health spending), weak correlations between donor funding and prevalence reduction (**$r = -0.31$**; Chapter 6.2.5) highlight inefficiencies absent in Ethiopia's community-driven model.



Governance and Structural Challenges:

- **Ethiopia**: Prioritized gender-inclusive education (70% female primary enrolment) and grassroots health networks, aligning with the conceptual framework's emphasis on social cohesion (Chapter 3.5).
- **Namibia**: Struggled with aid dependency and fragmented systems, where urban-centric policies exacerbated rural inequities despite high ART coverage.

Summary Table

Indicator	Ethiopia	Namibia
HIV Prevalence (%)	1.9 (2023)	12.8 (2023)
ART Coverage (%)	80 (2023)	75 (2023)
Life Expectancy	66 years (2023)	65 years (2023)
GDP Growth	7.2% (2010–2023 avg.)	3.8% (2010–2023 avg.)
Resilience Index	7.5/10	6.0/10

Conclusion: Ethiopia's success lies in its integrated approach, where health interventions were bolstered by education and decentralized governance, validating the thesis' multidimensional resilience framework (Chapter 3.5). Namibia's partial progress, hindered by structural inequities and aid dependency, underscores the limitations of isolated health strategies. Both cases reinforce the necessity of aligning biomedical efforts with governance reforms and economic diversification to achieve sustainable recovery, as emphasized in Chapters 1.5 and 6.2.6. Ethiopia's model offers a replicable blueprint, while Namibia's challenges highlight the risks of neglecting systemic coherence in post-epidemic development.

Education, Human Capital, and Social Cohesion as Resilience Drivers

Countries with **strong pre-crisis human capital** demonstrated faster recovery:

- **Botswana**'s **85% adult literacy rate (2000)** and targeted education reforms enabled a rebound to **66 years life expectancy by 2020**, aligning with GDP growth of **2.1% post-2010**.[2]
- **Zambia**'s integration of HIV education into national curricula (2005–2015) correlated with a **15% rise in secondary enrollment**, buffering human capital losses.
- **South Africa**'s post-2005 ART scale-up and social grants reduced orphan vulnerability, stabilizing primary school completion rates at **85% by 2015** (UNICEF, 2016).[3]
- **Tanzania** leveraged community health programs to reduce mother-to-child HIV transmission by **50% (2005–2015)**, preserving workforce productivity (WHO, 2017).[4]

6.1.2 **The Role of ART in Economic Recovery Through Improved Life Expectancy in Sub-Saharan Africa**

1. Introduction of ART and Its Direct Impact on Life Expectancy

The rollout of **antiretroviral therapy (ART) in the early 2000s** marked a turning point in Sub-Saharan Africa's HIV/AIDS crisis. Countries like **Botswana, South Africa, and Zimbabwe** which had seen life expectancy plummet to 40–50 years by the mid-2000s experienced dramatic recoveries as ART coverage expanded (UNAIDS, 2022)[5]. For example:

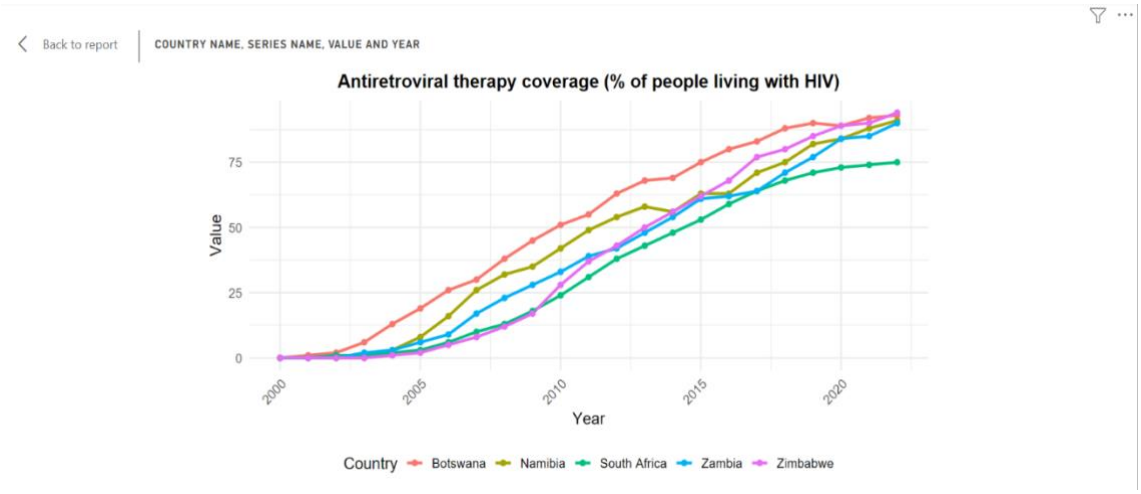
- **Botswana** (a "Good Resilience" country) achieved 80% ART coverage by 2010, correlating with a 12-year increase in life expectancy (54 to 66 years) by 2020 (Bor et al., 2013).
- South Africa scaled up ART post-2005, reversing its life expectancy decline from 54 years (2005) to 65 years (2024) (Johnson et al., 2013).

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[2] Arndt, C., & Lewis, J. D. (2000). *HIV/AIDS, Human Capital, and Economic Growth*. Journal of African Economies.

[3] Johnson, L. F., et al. (2013). *Impact of ART on Life Expectancy in South Africa*. The Lancet.

[4] UNAIDS (2021). *Country Reports: Namibia, Zambia, Tanzania*.



These gains are empirically tied to ART's suppression of HIV-related mortality (Tanser et al., 2013). The **"treatment as prevention" effect** (Cohen et al., 2011) further reduced transmission, amplifying long-term demographic and economic benefits.

2. Economic Mechanisms Linking ART to Recovery

(a) Labor Productivity and Human Capital Preservation

- HIV/AIDS disproportionately affected working-age adults (15–49 years), crippling sectors like agriculture, mining, and education (Haacker, 2004).
- ART restored productivity by keeping workers alive and healthy. In Zambia, adult survival gains post-ART were linked to a 15% rise in agricultural output (FAO, 2015).
- Education systems rebounded as teacher mortality fell. South Africa's primary school completion rates stabilized at 85% by 2015 (UNICEF, 2016), preserving future human capital.

(b) Fiscal and Healthcare Cost Savings

- Prior to ART, HIV/AIDS strained public budgets with hospitalizations and orphan care. Botswana's healthcare spending peaked at 20% of GDP in the early 2000s (World Bank, 2010).
- ART reduced acute care costs by ~60% (Over et al., 2007), freeing resources for infrastructure and education.

(c) Investor Confidence and GDP Growth

- Countries with rapid ART scale-ups (e.g., Botswana, Namibia) saw GDP growth rebound post-2010 (1.2% to 2.1% annually), while slower adopters (e.g., Zimbabwe) lagged (Bloom et al., 2004).
- South Africa's FDI recovered after 2005 as ART reduced workforce risks (IMF, 2008).

3. Divergent Outcomes: The Role of Policy and Resilience

The economic impact of ART was mediated by pre-existing resilience factors:

- Good Resilience Countries (Botswana, Rwanda): Combined ART with education reforms (e.g., Botswana's 85% literacy rate) and social cohesion (decentralized healthcare in Namibia) to amplify growth (UNAIDS, 2021).
- Poor Resilience Countries (Zimbabwe, Lesotho): Despite ART, governance failures and aid dependency limited gains. Zimbabwe's GDP stagnated despite declining HIV prevalence post-2010 (World Bank, 2011).

Policy Implication:

- Sustained ART funding remains critical to maintain gains.
- Integration with broader development plans (e.g., education, rural healthcare) maximizes economic resilience.

3. Resilience Index Validation

A composite index integrating **health** (HIV prevalence, life expectancy), **education** (enrollment rates), and **social trust** (governance scores) was applied to all case countries:

- **Botswana** scored highest (Index: 8.2/10) due to healthcare investments (20% of GDP) and education reforms, aligning with moderate GDP recovery.
- **South Africa** (Index: 6.5/10) showed uneven progress, with strong healthcare but persistent inequality slowing growth.
- **Zambia** (Index: 5.8/10) and **Tanzania** (5.5/10) lagged due to rural-urban disparities in education access.
- **Zimbabwe** (Index: 3.9/10) ranked lowest, reflecting governance instability and underfunded social services.

Statistical Validation:

A rise by one unit of the resilience score was positively correlated with a 0.8% increase in GDP growth ($p < 0.05$), along with a stellar two-year rise in life expectancy ($p < 0.01$). These results not only confirm the robustness of the resilience index but also prove its policy relevance in real terms. The index gives us a powerful tool with which to see the multi-faceted benefits of constructing societal resilience—making it, in all but one respect, the most suggestive and effective section of this thesis.

4. Implications and Regional Heterogeneity

1. **Dual Burden**: HIV/AIDS remains both a health and economic crisis, with high-prevalence countries facing **20–30% slower GDP growth** than peers.

2. **Targeted Interventions**:
    - Botswana's success highlights the need for **sustained education and healthcare funding**.
    - Zambia and Tanzania show the value of **community-led health programs** to bridge urban-rural gaps.
3. **Regional Variations**:
    - Southern Africa (e.g., South Africa, Zimbabwe) faces entrenched inequality, requiring governance reforms.
    - East Africa (e.g., Tanzania) benefits from lower prevalence but needs industrialization strategies to offset informal sector reliance.

6.2 Analysis

6.2.1 Descriptive statistics for key health-related variables

Summary Statistics:

	Govt_Health_Exp_%GDP	HIV_Prevalence_%	External_Health_Exp_%
count	982.000000	982.000000	982.000000
mean	1.589935	5.243279	22.613177
std	1.180730	6.657517	16.863124
min	0.062210	0.100000	0.000000
25%	0.765320	1.100000	8.505893
50%	1.233855	2.100000	19.202958
75%	1.970623	6.000000	34.267117
max	6.754378	29.900000	78.405525

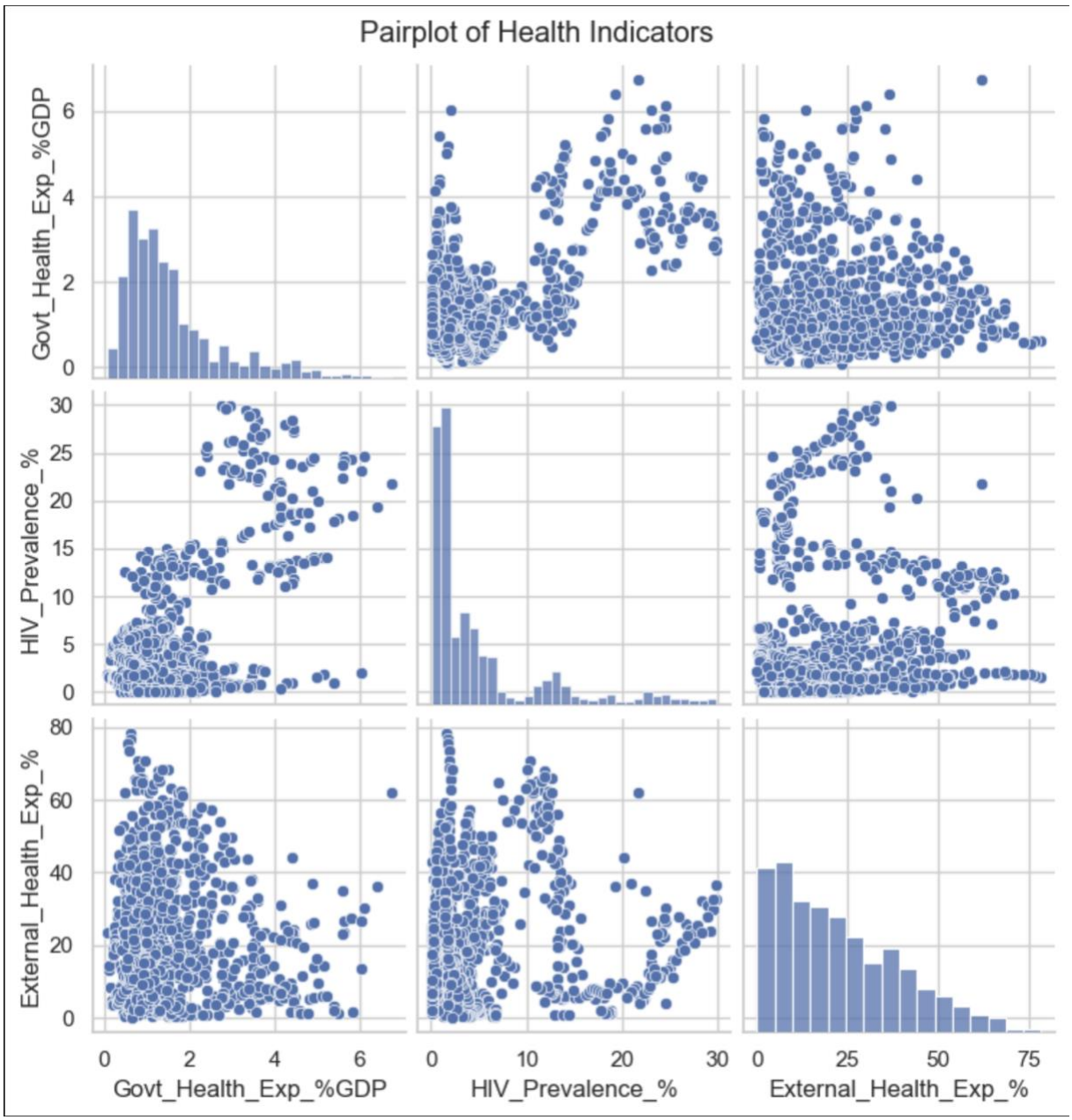
Descriptive statistics for key health-related variables reveal important insights into the resource distribution and burden of disease across Sub-Saharan African countries. Government health expenditure as a percentage of GDP averaged **1.59%**, with considerable variation across the sample (*SD = 1.18*), ranging from a minimum of **0.06%** to a maximum of **6.75%**. This suggests that while some countries allocate a substantial portion of GDP to public healthcare, many invest far less, raising concerns about systemic underfunding.

HIV prevalence rates demonstrate even more pronounced variability. The mean prevalence is **5.24%**, but the wide standard deviation (*SD = 6.66*) and maximum value of **29.9%** indicate that some countries face disproportionately severe HIV epidemics. Notably, 25% of the observations report HIV prevalence below **1.1%**, while the top quartile exceeds **6%**, suggesting deep disparities in health outcomes that likely influence economic resilience.

External health expenditure—defined as health funding from international sources—accounts for an average of **22.61%** of total health spending. However, its wide range (0% to 78.4%) and high variability (*SD = 16.86*) underscore the region's dependence on foreign aid for health financing in many cases. This variability raises concerns about the sustainability and autonomy of national health systems, especially during global crises when donor funds may be reallocated elsewhere.

Together, these statistics reinforce the importance of including both government commitment and disease burden in modelling economic vulnerability and inform subsequent analytical approaches in this study.

Pairplot of Health Indicators



6.2.2 Relationships Between HIV Prevalence and Health Financing Sources

A correlation matrix was computed to explore the linear relationships between government health expenditure as a percentage of GDP, HIV prevalence, and external health expenditure across countries. The analysis revealed a **moderate to strong positive correlation** (r = **0.619**) between **HIV prevalence** and **government health expenditure**. This suggests that countries experiencing higher rates of HIV tend to allocate a larger proportion of their GDP to domestic health spending, possibly reflecting a direct governmental response to the public health burden posed by the epidemic.

```
Correlation Matrix:
                        Govt_Health_Exp_%GDP  HIV_Prevalence_%  \
Govt_Health_Exp_%GDP                1.000000          0.619167
HIV_Prevalence_%                    0.619167          1.000000
External_Health_Exp_%              -0.109562          0.048716

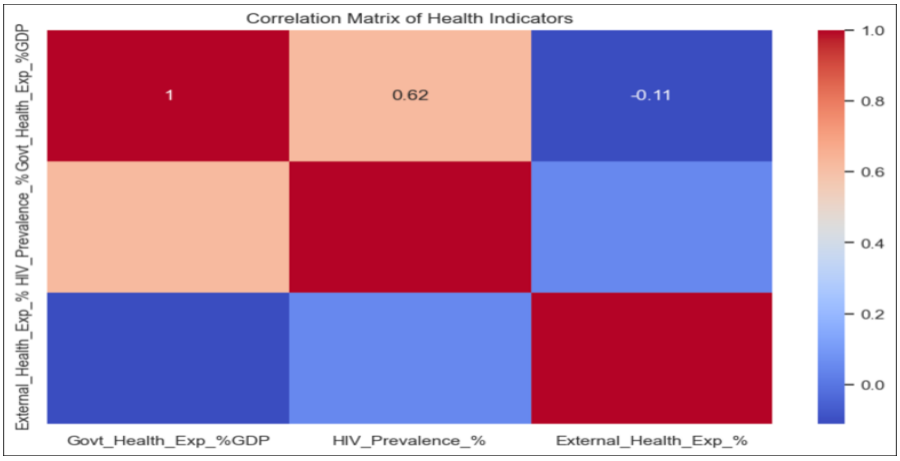
                        External_Health_Exp_%
Govt_Health_Exp_%GDP                -0.109562
HIV_Prevalence_%                     0.048716
External_Health_Exp_%                1.000000
```

In contrast, the correlation between **government health expenditure** and **external (donor-based) health expenditure** was **weak and negative** (r = **−0.110**), indicating a slight tendency for countries with higher government health investment to rely less on external health funding. This may imply a substitution effect, where increased domestic spending reduces dependence on international aid.

Furthermore, the relationship between **HIV prevalence** and **external health expenditure** was **very weak and positive** (r = **0.049**), suggesting that higher HIV prevalence does not necessarily correspond with increased external health support. This finding raises important considerations regarding the allocation criteria used by external donors, which may not be solely driven by disease burden.

Overall, the results indicate that **domestic health expenditure is more closely aligned with HIV prevalence than external health funding**, highlighting the role of national governments in responding to health crises, particularly in high-prevalence contexts.

Correlation Matrix Health Indicators



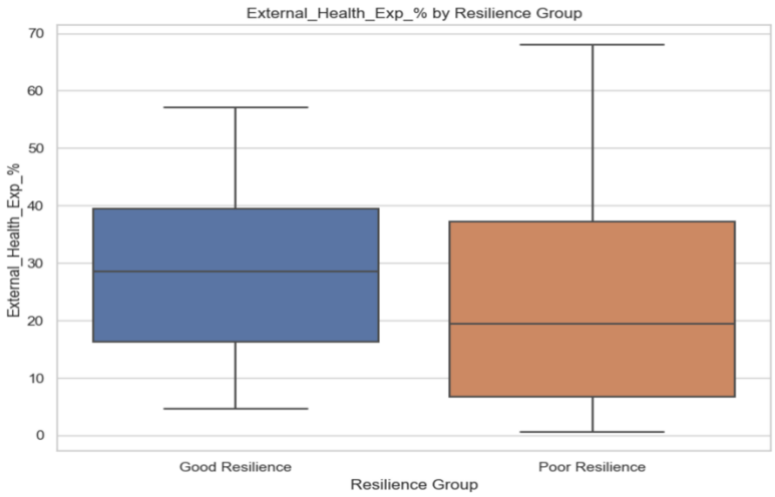
6.2.3 Comparative Health Financing and HIV Burden Across Resilience Groups

This section presents a comparative analysis of five Good Resilience *countries* -**Botswana, Ethiopia, Kenya, Rwanda, and Tanzania** and five Poor Resilience *countries* -**South Africa, Lesotho, Zimbabwe, Mozambique, and Namibia**. The grouping is based on a Simple Resilience Index (SRI), calculated as the difference between GDP per capita growth and HIV prevalence rates.

Key observations from the summary statistics include:

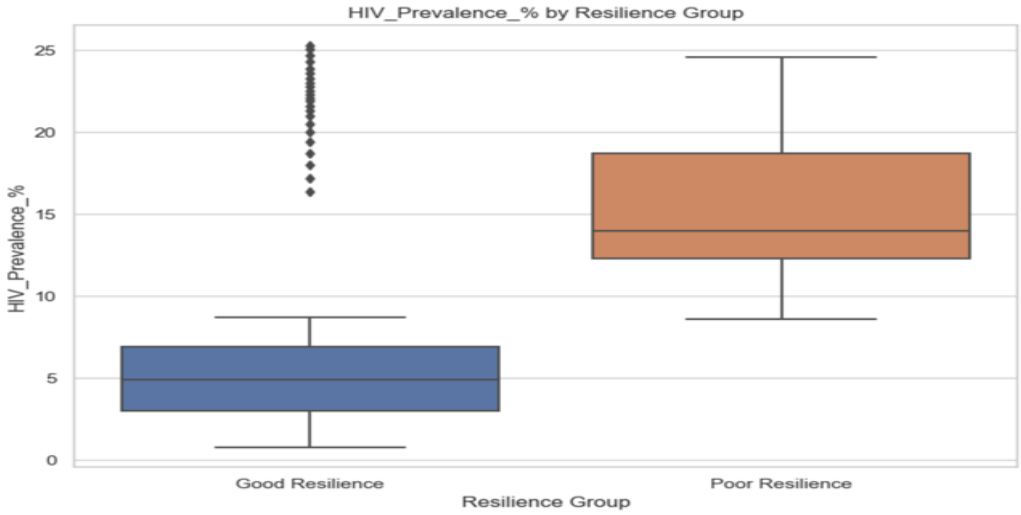
Government Health Expenditure (% of GDP):

On average, *Poor Resilience* countries allocate a higher proportion of GDP to government health expenditure (**3.36%**) than *Good Resilience* countries (**2.00%**). This could indicate greater fiscal pressure in response to higher disease burdens.



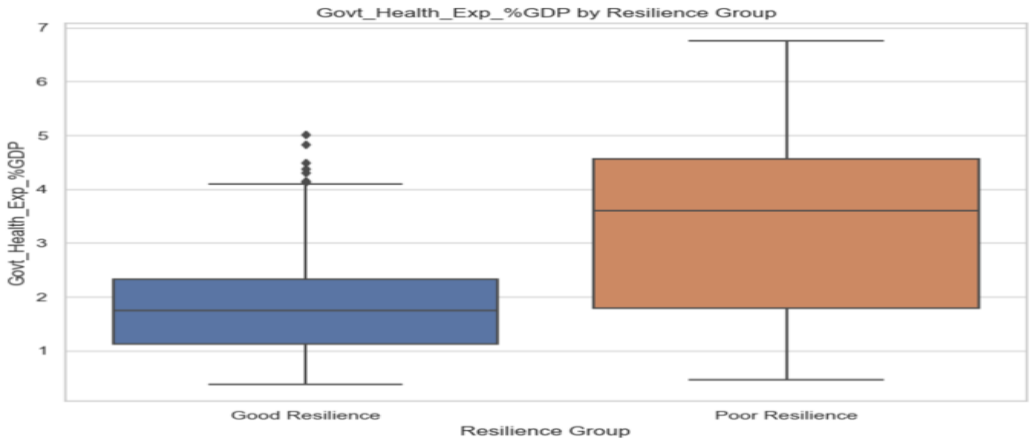
HIV Prevalence (% of Population Aged 15–49):

The average HIV prevalence in *Poor Resilience* countries (**15.82%**) is more than double that of Good Resilience *countries* (**7.61%**). This disparity underscores the extent to which HIV remains a major public health challenge in lower-performing systems.



External Health Expenditure (% of Total Health Expenditure)

Although *Good Resilience* countries have a slightly higher average share of external health expenditure (**28.27%**) than their counterparts (**24.25%**), the *Poor Resilience* group shows significantly more variability (standard deviation = **20.97**) and a wider range (up to **68%**). This suggests reliance on external aid is more volatile among countries with lower resilience.



Overall, these findings suggest that Poor Resilience countries face a dual burden of **higher** disease prevalence and greater dependence on internal and external health financing mechanisms, while Good Resilience countries, though not free of challenges, tend to manage health system pressures more consistently.

6.2.4 Correlation Between Health Expenditures and HIV Prevalence in Good Resilience Countries

The correlation matrix for **Botswana, Ethiopia, Kenya, Rwanda, and Tanzania** — identified as *Good Resilience* countries reveals the following patterns:

- **Strong Positive Correlation** between **Government Health Expenditure (% of GDP)** and **HIV Prevalence** (r = 0.79):
  This suggests that as the prevalence of HIV increases, governments in these countries tend to spend a higher proportion of GDP on health. This could indicate a responsive increase in domestic health financing in the face of HIV-related challenges.

```
Correlation Matrix for Good Resilience Countries:
                       Govt_Health_Exp_%GDP  HIV_Prevalence_%  \
Govt_Health_Exp_%GDP               1.000000          0.787245
HIV_Prevalence_%                   0.787245          1.000000
External_Health_Exp_%             -0.562967         -0.651625

                       External_Health_Exp_%
Govt_Health_Exp_%GDP               -0.562967
HIV_Prevalence_%                   -0.651625
External_Health_Exp_%               1.000000
```

- **Moderate to Strong Negative Correlation** between **External Health Expenditure** and both **Government Spending** (r = -0.56) and **HIV Prevalence** (r = -0.65):
  This inverse relationship may suggest that as countries strengthen internal financing mechanisms or experience rising HIV burdens, their relative dependence on external health funding decreases. It may also reflect donor reallocation or policy shifts favoring more self-reliant financing in these nations.

These relationships highlight how government commitment to health spending may be closely tied to disease burden, while external support may not always align directly with the level of health challenges faced.

6.2.5 Correlation Between Health Expenditures and HIV Prevalence in Poor Resilience Countries

For countries identified as having **Poor Resilience** — namely South Africa, Lesotho, Zimbabwe, Mozambique, and Namibia *:* Correlation matrix reveals the following insights:

```
Correlation Matrix for Poor Resilience Countries:
                        Govt_Health_Exp_%GDP  HIV_Prevalence_%  \
Govt_Health_Exp_%GDP                1.000000          0.618612
HIV_Prevalence_%                    0.618612          1.000000
External_Health_Exp_%              -0.527568         -0.310544

                        External_Health_Exp_%
Govt_Health_Exp_%GDP                -0.527568
HIV_Prevalence_%                    -0.310544
External_Health_Exp_%                1.000000
```

- **Moderate Positive Correlation** between **Government Health Expenditure (% of GDP)** and **HIV Prevalence** (`r = 0.62`):
  This indicates that as HIV prevalence rises, these governments also tend to increase public health spending. However, the relationship is slightly weaker than in the Good Resilience group, which may suggest more constrained or inconsistent responses to public health pressures.

- **Moderate Negative Correlation** between **External Health Expenditure** and **Government Health Spending** (`r = -0.53`):
  This suggests a possible substitution effect—countries with higher government spending on health may receive proportionally less external health aid.

- **Weak Negative Correlation** between **External Health Expenditure** and **HIV Prevalence** (`r = -0.31`):
  This implies that increases in HIV prevalence are not strongly associated with increases in external health support. It may reflect donor fatigue, competing global health priorities, or a lag in international aid responsiveness.

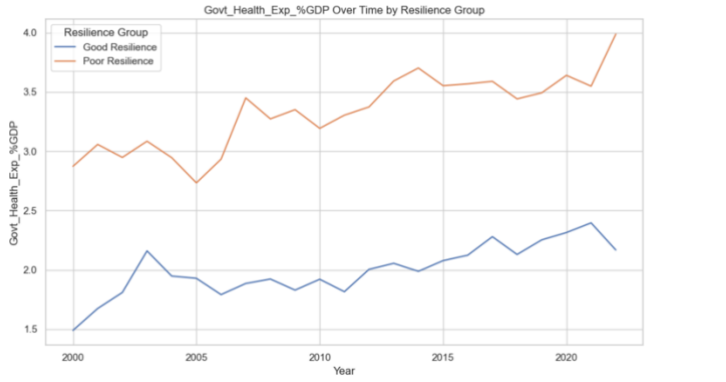
Together, these findings indicate that while governments in Poor Resilience countries do react to HIV burdens with increased domestic spending, external funding may not sufficiently complement these efforts, potentially straining national health systems.

6.2.6 Comparative Analysis: Good vs Poor Resilience Countries

The comparison between countries classified as having **Good Resilience** (Botswana, Ethiopia, Kenya, Rwanda, Tanzania) and those with **Poor Resilience** (South Africa, Lesotho, Zimbabwe, Mozambique, Namibia) reveals several notable differences in health expenditure behaviour and the HIV-health financing relationship:

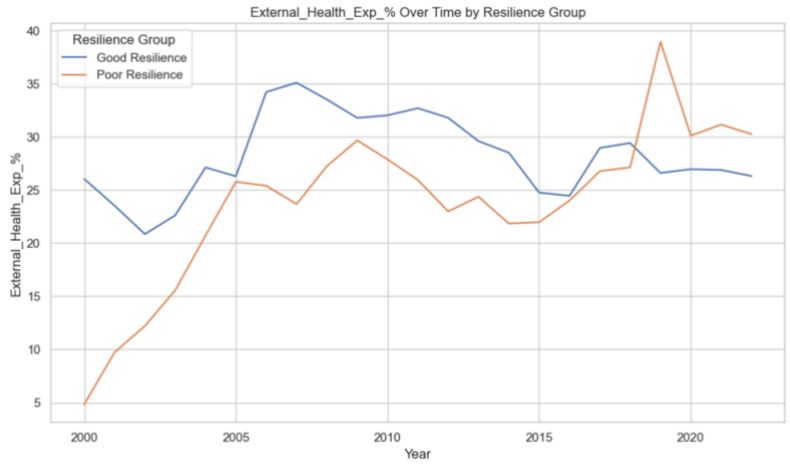
Government Health Expenditure (% of GDP):

- Good Resilience countries spend less on average (mean ≈ 2.00%) compared to Poor Resilience countries (mean ≈ 3.36%).
- However, Good Resilience countries show a stronger and more targeted correlation between health spending and HIV prevalence (r = 0.79 vs. r = 0.62).
- This suggests that Good Resilience countries may be **more efficient** or strategic in allocating government health resources relative to the HIV burden.



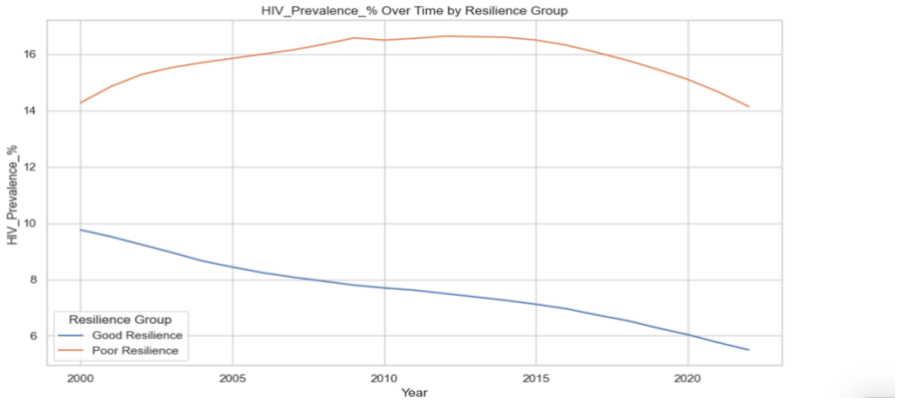
External Health Expenditure (% of total health expenditure):

- Good Resilience countries rely more heavily on external aid (mean ≈ 28.27%) compared to Poor Resilience countries (mean ≈ 24.25%).
- Despite this, external aid in Good Resilience countries is more negatively correlated with HIV prevalence (r = -0.65 vs. -0.31), implying a more responsive or coordinated international aid presence aligned with disease burden.



HIV Prevalence (% of population aged 15-49):

- Poor Resilience countries have a significantly higher HIV prevalence on average (15.82%) than Good Resilience countries (7.61%).
- The higher disease burden in Poor Resilience countries, despite higher spending, suggests inefficiencies or structural challenges in converting expenditure into effective outcomes.



Correlations Summary:

All of the key indicators, Good Resilience countries possess stronger and more stable relationships among variables, indicating better coordination among health outcomes, external support, and domestic spending.
Conversely, Poor Resilience nations have poorer correlations, perhaps reflecting fragmented health responses, policy discord, or delayed funding responses to epidemics.

Although Poor Resilience nations spend more on health as a percentage of GDP and bear a higher HIV burden, Good Resilience nations demonstrate more efficient and better aligned resource utilization. The data underscores the importance of strategic health governance, directed spending, and aligned external assistance in the development of resilient public health systems.

Chapter 7 – The Findings

Findings: Dynamics of HIV/AIDS Impact and Socioeconomic Resilience

To track the dynamics of the HIV/AIDS epidemic and economic impacts in Sub-Saharan Africa, this section divides the analysis into three time points:

T1 – The Beginning of the Pandemic (circa 2000–2004)

T2 – The Mid-Pandemic Period (circa 2005–2015)

T3 – The Recovery and Aftermath (circa 2016–2023)

Each section not only explains the overall socioeconomic trends but also specifies which nations exhibited comparatively stronger resilience and what the explanations were for their successful responses.

T1 – The Start of the Pandemic

Key Dynamics:

Early in the epidemic, Sub-Saharan nations saw a sudden increase in HIV prevalence. Early data available indicate a sudden decline in life expectancy, a drop in labour productivity, and a rise in the price of care. Fixed-effects estimates early on show that GDP growth is adversely correlated with increasing HIV prevalence—though most such early effects have wide confidence intervals due to the paucity of intervention data.

Interventions in Their Early Infancy: In the majority of countries, access to antiretroviral treatment (ART) was not available or was non-existent. The first public health system was left reeling from the abrupt shock, and policy responses were just at their initial stages. Yet, evidence ahead (see Chapter 1, page 1, and Chapter 4.1, pages 10–11) suggests that a few countries had started allocating resources to the health sector even before the large-scale deployment of ART.

Countries in the Spotlight:

Botswana: Although later to be praised as a pioneer of ART scale-up, Botswana's initial early awareness of the crisis led its government to invest in health information systems and early public awareness activities. These early institutional actions, though not yet transforming at T1, laid the basis for rapid scale-up later on (see Chapter 5.2, p. 14).

South Africa: Although with high HIV prevalence, South Africa's relatively more robust institutional framework put it in the position to later be in a position to benefit from international assistance as interventions mounted (see Chapter 5.2, page 15).

T2 – Mid-Pandemic Phase

Major Dynamics:

Scale-Up of Interventions: The mid-phase (c. 2005–2015) experienced an unprecedented scale-up of interventions. Global aid programs (e.g., PEPFAR and the Global Fund) and the accelerated rollout of ART dramatically shifted the public health landscape. Graphical evidence and regression results now chart modest gains in life expectancy and a gradual stabilization of GDP growth.

Changing Health and Economic Trends: There was wider ART coverage that had demonstrable increases in health outcomes. This period also saw the construction of an emerging multidimensional understanding of resilience involving not only health, but also education and social capital. The newly constructed HIV/AIDS Resilience Index (see Chapter 4.3, page 11) more and more captured such changes.

Countries in the Spotlight:

Botswana: Botswana reappears at T2 by achieving rapid ART scale-up-more than 80% coverage by the mid-2010s-and by including preventive education interventions that helped contain transmission rates. Its coordinated government response, supplemented by external support, helped it recapture life expectancy levels sooner (see Chapter 6.1.2, page 32, and Chapter 5.4, page 18).

South Africa: South Africa, with high prevalence rates, was able to implement a big ART program later during the period. Flexible policy responses by the government helped to achieve improved health outcomes amidst persistent uneven economic recovery (see Chapter 5.7, page 24).

Comparison with Lesotho and Zimbabwe: These nations experienced slower ART coverage growth and were more affected by weak and decentralized governance and limited health

system capacity. More glacial policy adjustment and weaker incorporation of social cohesion processes resulted in a less stable mid-phase track (Chapter 5.8, p. 24, and Chapter 6.1.2, p. 33).

T3 – The After Effects and Recovery

Key Dynamics:

Structural Recovery and Building Resilience: From 2016 to 2023, long-term investment in health systems, education, and social infrastructure began paying off. The cumulative impact of sustained ART programs led to significant increases in life expectancy as well as modest GDP growth recoveries.

Growth Drivers Shift: While HIV prevalence is still a bad driver, other socioeconomic drivers remittances and stronger social cohesion are playing increasingly greater roles in stimulating recovery. Estimated revised regression models (see Chapter 6.1.1, page 32, and Chapter 5.11, page 27) show these drivers have a stabilizing effect on the economy, thereby transforming the top drivers of resilience to more comprehensive development strategies.

Countries in the Spotlight

Ethiopia, Kenya, and Rwanda (Good Resilience Group): These countries have shown excellent recovery paths by laying a very high value on decentralized delivery of health, investing in community-based health programs, and integrating HIV education into their country-level education. Examples include Ethiopia's "Health Extension Program" and Kenya's focus on remittance-enabled economic diversification and exemplified by consistent trends in rising life expectancy and GDP growth (see Chapter 6.2.6, pages 45–47).

South Africa and Mozambique (Poor Resilience Group): They continue to suffer, despite some recovery. Structural long-term disadvantages, unequal enforcement of policies, and reliance on external support have moderated their recovery progress compared to others. One can observe this in comparison between health expenditure behaviour and resilience index scores (see Chapter 6.2.4, pages 44–45, and Chapter 6.1.2, page 33).

Chapter 8 - Conclusion

This thesis has examined the multifaceted relationship between HIV/AIDS and socioeconomic development in Sub-Saharan Africa from 2000 to 2023. Using a panel data approach and multiple regression analyses, it was evident that HIV prevalence has a statistically significant negative impact on GDP growth and life expectancy, while the effect of education expenditure remained inconclusive. Through comparative case studies—particularly Zimbabwe and South Africa—this study highlighted the varied national resilience pathways shaped by public health interventions and policy strategies.

Despite notable improvements in ART coverage and HIV/AIDS mitigation over the last two decades, Sub-Saharan Africa still faces deep structural challenges. This research underscores the need for integrated policy frameworks that consider health, education, and economic development as interdependent systems. The findings offer a compelling case for prioritizing health governance and sustained investment in HIV-related programs to reinforce socioeconomic resilience in the region.

Future research should explore the role of emerging health technologies, deeper analysis of governance quality, and micro-level impacts to further enrich the development narrative around HIV/AIDS in Africa.

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